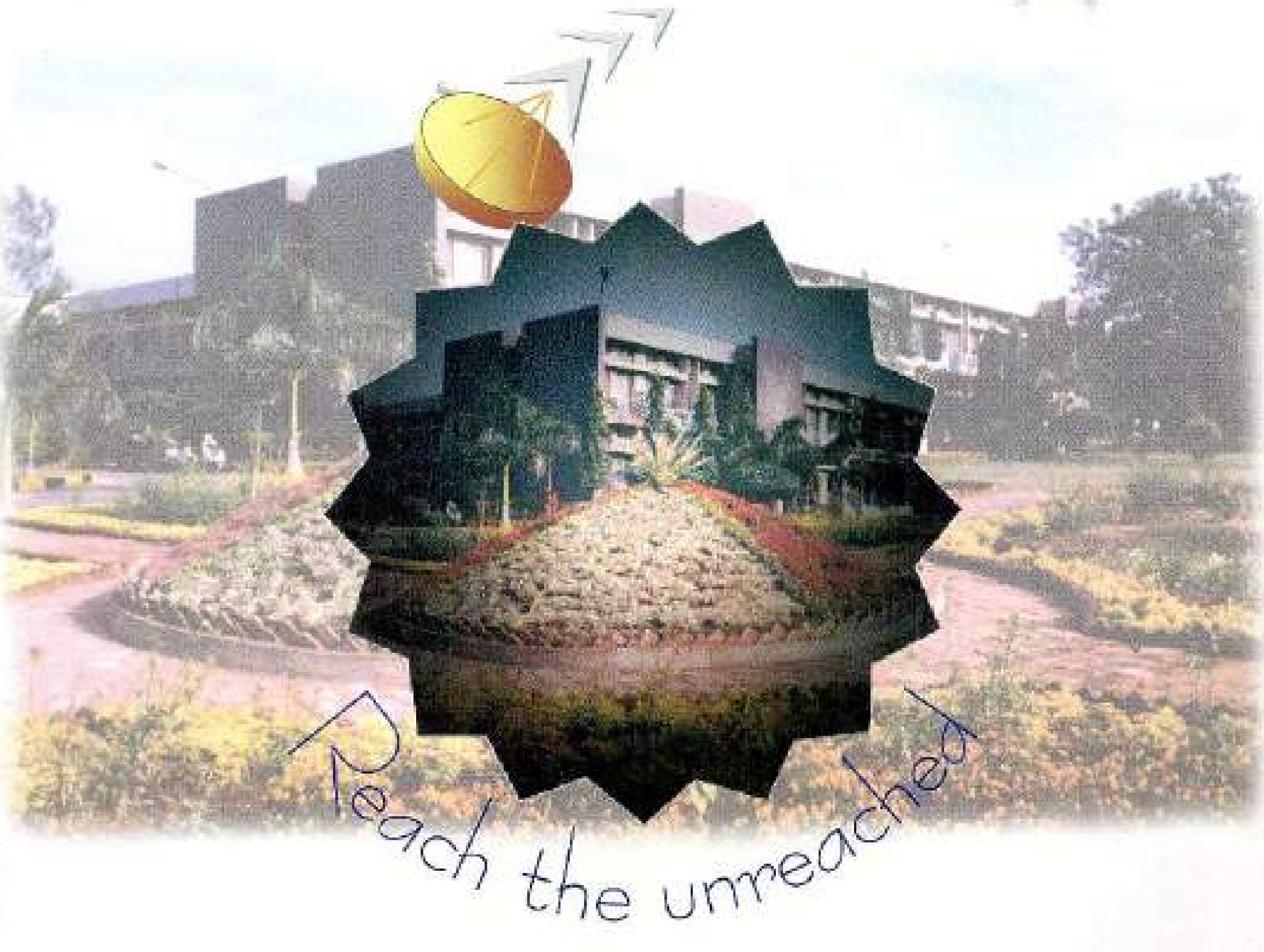
वार्षिक प्रतिवेदन
Annual Report 2001-2002



राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, राजेन्द्रनगर, हैदराबाद-५०० ०३०, भारत

National Academy of Agricultural Research Management

Rajendranagar, Hyderabad-500 030, India

http://icar.naarm.ernet.in

वार्षिक प्रतिवेदन
Annual Report 2001-2002

राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी, राजेन्द्रनगर, हैदराबाद-५०० ०३०, भारत

National Academy of Agricultural Research Management

Rajendranagar, Hyderabad-500 030, India

http://icar.naarm.ernet.in

Citation

NAARM 2001. Annual Report of the National Academy of Agricultural Research Management for 2001-2002 NAARM, Rajendranagar, Hyderabad, A.P., India

Credits

Compilation and Editing	:	Dr D. Rama Rao Dr B.S. Sontakki Mr R.V.V.S. Prakasa Rao
Cover Design & Layout	:	Mr P. Namdev
Photography	:	Mr L. Venkateswarlu

Published by Director, NAARM

Phone : 040-4015394, 4015395

Fax : 040-4015912

E-mail : naarm.hyd@naarm.ernet.in

Grams : NAARM

Contents

Preface

The Academy has completed 25 years of dedicated service to the nation. During these years the Academy has made significant contribution by imparting training to scientists, teachers, administrators and policy makers of the Indian National Agricultural Research System (NARS), and also those from other developing countries in Asia and Africa. In this process the Academy has been able to bring about changes in attitude, value system, and commitment in agricultural scientists. The international collaboration in research and training programmes with ISNAR, Tanzania and Yemen provided excellent experience, which will show a positive impact on our performance in the years to come. During this period, 45 programmes were held, and a total of 1028 scientists, teachers, administrators and finance officers were trained. The Academy also conducted two off-campus programmes one for the top executives of ICAR at IIM, Ahmedabad and the other at Postgraduate Institute of Agriculture, University of Peradeniya, Sri Lanka.

Under the policy support relating to management of agricultural research and education systems, the Academy organized one-day interactive meeting to review the performance appraisal of ICAR scientists, and prepare a new methodology in conformity with the project based budgeting. Utilization of funds for maintaining and creating infrastructure facilities for effective conduct of training programmes has been accomplished satisfactorily.

I wish to take this opportunity to put on record our sincere thanks and gratitude for the support, guidance, and encouragement received from Dr R.S. Paroda, Former Director General, ICAR; Dr Panjab Singh, Director General, ICAR, and Secretary, DARE, Govt. of India, Dr J.C. Katyal, DDG (Education). I specifically owe a gratitude to my colleagues Drs T. Balaguru, P. Manikandan, D. Rama Rao, Jagannadham Challa, and B. S.Sontakki who supported in preparation of this report. The assistance of Mr R.V.V.S. Prakasa Rao and Mr P. Namdev in layout design and printing is highly appreciated. I compliment all faculty members, technical, administrative and supporting staff as well as trainees for their contribution to the Academy.

Hyderabad
17-8-2002

(S.N. Saha)

1. Executive Summary

National Academy of Agricultural Research Management (NAARM), the premier Institute of the Indian Council of Agricultural Research entrusted with the responsibility to achieve higher levels of professional management in agriculture research and education, has been contributing to the Indian National Agricultural Research System (NARS) for nearly two and a half decades. The multi-dimensional programmes of training, research and consultancy received keen attention during the year 2001-2002, along with emphasis on infrastructure development.

The Academy's year-long Silver Jubilee Celebrations started with the NAARM Foundation Day lecture by Dr. M. Mahadevappa, Chairman, ASRB, New Delhi on September 1, 2001.

In the Academy's efforts to improve the foundation course based on the participants' feedback, the Foundation Course for Agricultural Research Service (FOCARS) was reorganized in a modular approach for the first time. It resulted in tangible gains in the form of learning outcomes. The Field Experience Training was organized on the principles of 'seeing is believing' and 'learning by doing' through multi-disciplinary participatory approach.

As a part of the HRD activities, a wide variety of training programmes were organized by the Academy for the benefit of scientists / faculty from ICAR and State Agricultural Universities (SAUs). During the reported period, 45 such programmes were held, and a total of 1028 scientists, teachers, administrators and finance officers participated in these programmes. The Academy also conducted two off-campus programmes viz. one Retreat Programme for Top Executives of ICAR at IIM, Ahmedabad (eighteen senior level functionaries attended the programme) and the other at Postgraduate Institute of Agriculture, University of Peradeniya, Sri Lanka (thirty-nine scientists from various research establishments in Sri Lanka attended the programme and got benefited).

Two Summer Schools, one on Recent Advances in Instructional Technology and another on Recent Advances in Agricultural Research Project Management were organized. In all, fortysix scientists from ICAR institutes and SAUs participated and got benefited.

Under the AHRD project, extensive training was imparted to sensitize the Professors, Associate Professors and Assistant Professors (numbering 162) of TNAU on Computer Aided Instructional Technology. In addition, 109 administrative functionaries were also given training on administrative management and computer applications. During the period, Senior Executives Computer Lab and Executive Committee Room were developed.

A National Seminar on "Veterinary Drugs and Pharmaceuticals" was organized in collaboration with Department of Science and Technology, New Delhi, to assess the status of R & D in the area of veterinary drugs and pharmaceuticals. Inaugurated by Padma Vibhushan Prof. M.M. Sharma, the Seminar was attended by 64 participants from ICAR institutes, SAUs, Veterinary Drug and Pharmaceuticals Industry and the Government institutions.

One of the significant achievements of the year was the Pilot Distance Training Module on the theme "Focusing Agricultural Research on Poverty Alleviation", under the ISNAR-ICAR / NAARM collaborative project, which was prepared and released for implementation in five SAUs - MPKV, Rahuri, ANGRAU, Hyderabad; UAS, Dharwad, Karnataka; TNAU, Coimbatore and TANUVAS, Chennai. Online Directory of National Agricultural Research System was developed and published on NAARM Web Site (http://icar.naarm.ernet.in) for the benefit of all the institutions and scientists working along the length and breadth of the country. It is a kind of ready reckoner for accessing the research stations through e-mail, telephone and fax. It is useful for those working in the NARS in India and abroad, and other agencies involved in agricultural development in India.

Under the NATP sub-project on 'Institutionalization of Priority Setting, Monitoring and Evaluation, and Networking of Social Scientists', the Strategic Research and Extension Plan (SREP) guidelines were revised by strengthening the research component, and the revised guidelines are currently being utilized for developing effective SREPs. Under the Human Resource Development sub-project, a variety of computer-based instructional materials on Gender were developed. A case study exercise on characterization and prioritization of research in rainfed rice production has been developed using the district level databases in ARCVIEW GIS.

Under the A.P. Cess Fund Research Scheme on 'Developing Instructional Modules for Gender in Agricultural Curriculum', three instructional modules were prepared on gender roles, gender resource management and gender time management both in print and also as CD. The developed modules were demonstrated to the teachers of TNAU, Coimbatore and ANGRAU, Hyderabad.

In order to understand the state-of-the-art of the implementation of Farming Systems Research (FSR) approach in India, a case study was developed based on the primary and secondary information collected on the Integrated Farming Systems Development Programme undertaken by the Agricultural Research Station, Aruppukottai of TNAU, Coimbatore. The productivity, profitability and sustainability of the integrated farming systems developed by the scientists in the research station as well as those tested at the farmers field were analyzed and described in the case study.

Under the policy support relating to management of agricultural research and education systems provided to ICAR by the Academy, a one-day meeting was organized to review the performance appraisal of ICAR scientists and to prepare a new methodology in conformity with the project based budgeting. The Academy developed training needs assessment for ICAR employees which was approved and is being implemented.

The Academy has completed the major part of production of video film on Hybrid Rice Cultivation for the Directorate of Rice Research, Hyderabad. Another video on National Research Centre for Agro-forestry, Jhansi is under production.

Under NATP, one faculty member underwent three-months advanced training in HRD at the Ohio State University, USA, and another faculty member undertook three-week study visit to ISNAR, the Netherlands, on performance assessment of agricultural research organizations.

The Library of the Academy was put in the path of digitization for the first time. In order to computerize the library activities and to facilitate web online public access catalogue of library resources, LIBSYS package was procured.

The faculty members published 28 scientific papers and presented 23 papers in various workshops, seminars, symposia, etc. During the year, three bulletins viz. 'NAARM Achievements Profile', 'AHRD Project Document', and 'Rajbasha Karyanvayan Pachees Varsh Ki Pragathi Sameeksha' were released.

The Academy won the Rajbhasha Puraskar of Town Official Language Implementation Committee, South Central Railway, for the year 1999-2000 for the best work done in the implementation of the Official Language in twin cities.

Dr J.C. Katyal, Director, received the Lifetime Achievement Award of S. S. Ranade Memoral Trust, Pune, for his long-standing research contribution on micronutrients in agriculture.

The NAARM Sports contingent secured five individual and one team event prizes in the ICAR Inter-Institutional Tournament held at CIFT, Cochin; and in the final Inter-Zonal Tournament held at IARI, New Delhi, the Academy secured five individual prizes, including the overall championship for women.

The Academy bagged the Hyderabad Rose Society Challenge Cup, Deccan Chronicle Trophy and T. Chandrasekhar Reddy Memoral Tropy at the 26th Annual Rose Show held at the Horticulture Centre, Hyderabad. Apart from the above, the Academy also won 13 first prizes, 15 second prizes and 9 third prizes under different sections of flower display.

Activities at a Glance

April 2001

- A meeting-cum-workshop to bring Uniformity in Course Curricula and Syllabi at postgraduate level in the discipline of Seed Science and Technology was organized from April 6 to 7, 2001. Thirty-four members participated in the programme. Drs S.N. Saha, A. Gopalam and N. Hanumantha Rao coordinated the programme.
- A three-day workshop on Motivation Techniques was organized from April 10 to 12, 2001. Four members attended the workshop. Drs M.M. Anwer and Janardan Jee coordinated the workshop.
- A sensitization training programme on Computer Aided Instructional Technology for TNAU teachers was organized from April 10 to 12, 2001. Eighteen teachers from Tamil Nadu Agricultural University, Coimbatore participated in the programme. Drs A. Gopalam, S.K. Soam and R.V.S.Rao coordinated the programme.
- A one-week training programme on Designing and Developing Web Pages was organized from April 16 to 21, 2001. Twelve scientists from Agricultural Technology Information Centres participated in the programme. Drs N. Sandhya Shenoy, and M.N. Reddy coordinated the programme.
- A two-week programme on Training of Trainers was organized from April 17 to 28, 2001. Drs T. Balaguru and P. Manikandan coordinated the programme.
- A Summer School on Recent Advances in Agricultural Research Project Management was organized from April 18 to May 8, 2001. Twenty-one participants attended the programme. Dr J. C. Katyal was the Summer School Director. Drs T. Balaguru, R. Kalpana Sastry, V.K. Jayaraghavendra Rao and Bharat S. Sontakki coordinated the programme.
- Fifth Management Development Programme in Agricultural Research was organized from April 19 to 25, 2001. Six heads of the Divisions of ICAR institutes participated in the programme. Drs Jagannadham Challa and S.N. Saha coordinated the programme.
- Computer Based Instructional Technology for the TNAU faculty was organized from April 26 to May 2, 2001. Twenty-one faculty members participated in the pogramme. Drs K. Vidyasagar Rao, M.M. Anwer and R. Kalpana Sastry coordinated the programme.

May 2001

- A Summer School on Recent Advances in Instructional Technology was organized from May 10 to 30, 2001. Twenty-five Assistant Professors and Associate Professors of State Agricultural Universities participated in the pogramme. Dr A. Gopalam coordinated the Summer School.
- A three-day training programme on Modern Management of Administration was organized for the Administrative and Accounts Offices of Tamil Nadu Agricultural University, Coimbatore, from May 17 to 19, 2001. Twenty-six members attended the programme. Mr M. Suresh Kumar, Mr B. Pandu Reddy and Dr N. Sandhya Shenoy coordinated the pogramme.
- A three-day sensitization training programme on Computer Aided Instructional Technology for Professors of TNAU, Coimbatore was organized from May 28 to 30, 2001. Twenty-eight members attended the pogramme. Drs T. Balaguru, S.K. Nanda and B.S. Sontakki coordinated the programme.

- A one-day workshop on Awareness of Intellectual Property Rights in Indian Agriculture was organized on May 31, 2001 in collaboration with the Patent Facilitating Centre (PFC) of Technology Forecasting and Assessment Council (TIFAC), Department of Science and Technology, New Delhi. Dr R. Kalpana Sastry coordinated the workshop, which was attended by sixty-three scientists of Indian NARS.

June 2001

- A ten-day programme on Agricultural Scientist Development for Personal and Organizational Effectiveness was organized from June 12 to 23, 2001. Eight senior scientists of ICAR institutes participated in the programme. Drs S. Shanmugam and Jagannadham Challa coordinated the programme.
- A one-week training programme on Instructional Aids for Effective Presentation was organized from June 18 to 23, 2001. Drs N. Sandhya Sehenoy and R. Kalpana Sastry coordinated the programme.
- A training programme on Agricultural Research Prioritization Techniques was organized from June 25 to 30, 2001. Twenty-seven scientists participated in the programme. Drs K.P.C. Rao and B.S. Chandel coordinated the programme.
- A Faculty Development Programme in Computer-based Instructional Technology for faculty members of Tamil Nadu Agricultural University, Coimbatore was organized from June 26 to July 2, 2001. Thirty-two faculty members attended the programme. Drs N. Hanumantha Rao and K.M. Reddy coordinated the programme.

July 2001

- A five-day workshop on Gahan Hindi Prashikshan was organized from July 2 to 6, 2001. Forty one members attended the workshop. Dr A. Gopalam and Mr Pradeep Singh coordinated the workshop.
- A refresher course on Information Technology in Agriculture was organized from July 10 to 30, 2001. Twenty-eight scientists of ICAR institutes and SAUs participated in the programme. Drs N. Hanumantha Rao and M. Narayana Reddy coordinated the programme.
- A one-week programme on Organizational and Management (O&M) Reforms in Administrative and Financial Management was organized under NATP from July 23 to 31, 2001. Thirty-four administrative and financial personnel of ICAR institutes participated in the programme. Mr M. Suresh Kumar, Mr S.K. Pathak and Dr B.S. Sontakki coordinated the programme.

August 2001

- A ten-day programme on Analytical Tools and Techniques for Assessing Agricultural Sustainability was organized from August 1 to 10, 2001. Twelve scientists of ICAR institutes and SAUs participated in the programme. Dr M. Narayana Reddy and Dr N.Hanumantha Rao coordinated the programme.
- A one-week programme on Computer Applications in Agricultural Research and Education for Administrative Staff of Tamil Nadu Agricultural University, Coimbatore was organized from August 2 to 8, 2001. Twenty-one members attended the programme. Dr M. Narayana Reddy, Mr Suresh Kumar and Mr S.K. Pathak coordinated the programme.
- A two-week programme on Computer Applications in Agriculture was organized from August 20 to 31, 2001. Eighteen scientists from ICAR institutes attended the programme. Drs K.M. Reddy and K. Vidyasagar Rao coordinated the programme.

September 2001

- A refresher course on Recent Advances in Agricultural Research Project Management was organized from September 10 to 30, 2001. Nineteen members attended the programme. Drs D. Rama Rao and M.M. Anwer coordinated the course.

- A one-week training programme on Impact Assessment of Agricultural R & D was organized from September 17 to 22, 2001. Drs B.S.Chandel and K.P.C.Rao coordinated the programme.

October 2001

- A training programme on Stress Management was organized from October 17 to 24, 2001. Four scientists from ICAR institutes participated in the programme. Drs S. Shanmugam and P. Manikandan coordinated the programme.
- Training workshop on the Modalities of Implementation of Pilot Distance Training Programme was organized under the guidance of the experts from Commonwealth of Learning, Canada and ISNAR, The Hague, The Netherlands from October 24 to 26, 2001. The designated Site Coordinators from 14 Learning sites and five Nodal Officers from the five Sate Agricultural Universities participated along with NAARM Project team.
- A ten-day training programme on GIS Applications in Agricultural Research was organized from October 30 to November 9, 2001. Eight members attended the programme. Drs N. Hanumantha Rao and M. Narayana Reddy coordinated the programme.

November 2001

- A four-day workshop on Prioritization and Obligations of Official Language Policy in ICAR was organized from November 6 to 9, 2001. Forty-seven members attended the workshop. Dr A. Gopalam and Mr D. Venkateswarlu coordinated the workshop.
- Sensitization training programme on Computer Aided Instructional Technology for Professors of TNAU, Coimbatore was organized from November 8 to 10, 2001. Thirty-five professors attended the programme. Drs P. Manikandan, M.Narayan Reddy and S.K. Soam coordinated the programme.
- A ten-day programme on Advanced Data Analytical Techniques for Agricultural Research and Management was organized from November 19 to 29, 2001. Two members attended the programme. Drs M. Narayana Reddy and K. Vidyasagar Rao coordinated the prorgramme.
- A one-week programme on Computer Based Instructional Technology for Assistant Professors and Associate Professors of TNAU, Coimbatore was organized from November 19 to 25, 2001. Twenty-eight faculty members attended the programme. Drs K.P.C. Rao, K. Hanumantha Rao and A. Gopalam coordinated the programme.
- A one-week Management Development Programme in Agricultural Research was organized from November 21 to 27, 2001. Fourteen heads of the Divisions of ICAR institutes participated in the programme. Drs Jagannadham Challa and R.V.S. Rao coordinated the programme.

December 2001

- A Refresher Course on Information Technology in Agriculture was organized from December 3 to 23, 2001. Twenty-seven senior scientists from ICAR institutes and SAUs attended the programme. Drs K. Vidyasagar Rao, B.S. Sontakki, and S.K. Soam coordinated the programme.
- A one-week programme on Development Oriented Research in Agriculture was organized from December 10 to 15, 2001. Ten members attended the programme. Drs S.K. Soam and B.S. Sontakki coordinated the programme.

- A one-week sponsored programme on Computer Applications in Agricultural Research and Education was organized from December 10 to 16, 2001. Twenty-eight Administrative and Accounts Officers of TNAU, Coimbatore attended the programme. Drs R. Kalpana Sastry, N. Hanumantha Rao and B.S. Chandel coordinated the programme.

- A four-day Executive Development Programme on Agricultural Research Management was organized from December 21 to 24, 2001. Fourteen members attended the programme. Drs Jagannadham Challa and S.N. Saha coordinated the programme.

- A three-day sponsored programme on Educational Psychology and Leadership Development was organized from December 27 to 29, 2001. Fifty-four Assistant and Associate Professors of TNAU, Coimbatore attended the programme. Drs A. Gopalam and S.N Saha coordinated the programme.

January 2002

- A one-week Refresher Course on Administrative and Financial Management was organized from January 16 to 24, 2002. Twenty-five Administrative Officers and Finance and Accounts Officers of ICAR institutes participated in the programme. Mr M. Suresh Kumar and Mr S. K. Pathak coordinated the programme.

- A three-day workshop on Methodologies for Manpower Planning in Agriculture was organized from January 17 to 19, 2002. Fourteen Professors and Associate Professors of SAUs and ICAR institutes participated in the programme, which was coordinated by Drs D. Rama Rao and S.K. Nanada.

- A one-day meeting on Review of Performance Appraisal and Preparation of New Rules and Guidelines for ICAR was organized on January 25, 2002. Thirty Directors and Heads of the Divisions of ICAR institutes attended the programme. Dr D. Rama Rao coordinated the programme.

February 2002

- A five-day workshop on Gahan Hindi Prashikshan was organized from February 4 to 8, 2002. Forty members attended the workshop. Dr A. Gopalam and Ms J. Renuka coordinated the workshop.

- A ten-day programme on Computer Applications in Agriculture was organized from February 5 to 15, 2002. Seven members attended the programme. Drs K. Vidyasagar Rao and K.M. Reddy were the coordinators.

March 2002

- A five-day programme on Project Management Using Microsoft Project was organized from March 4 to 8, 2002. Six members attended the programme. Drs S.K. Nanda, N.H. Rao and D. Rama Rao coordinated the programme.

- A three-day workshop on Change Management was organized from March 14 to 16, 2002. Eight members attended the programme. Drs S.N. Saha, M.M. Anwer, B.S. Sontakki coordinated the programme.

- Sponsored by the Department of Science and Technology, New Delhi, a two-day National Seminar on Veterinary Drugs and Pharmaceuticals was organized on March 23 and 24, 2002 in association with Acharya N.G. Ranga Agricultural University, Hyderabad. Seventy-five scientists, academicians and private sector executives working in the field of veterinary pharmacology, toxicology and medicine attended the seminar. Drs D. Rama Rao, NAARM, and K.S. Reddy, ANGRAU, coordinated the Seminar.

Introduction

The National Academy of Agricultural Research Management (NAARM) was established by the Indian Council of Agricultural Research (ICAR) at Hyderabad in 1976, to fulfil an important need to address issues related to agricultural research and education management. In the initial years, it primarily imparted foundation training to the new entrants of the Agricultural Research Service of ICAR. Subsequently, its role expanded to include training of all categories of scientists, faculty members, administration and financial personnel of Indian National Agricultural Research System (NARS), as well as those from other developing country NARS. The faculty members of the Academy are extensively trained across a range of areas covering agricultural research and education management, both in India and abroad. To improve the utility and credibility of training programmes, the Academy conducts research in the areas of agricultural research and education management. NAARM provides policy support in these areas to improve the efficiency and effectiveness of NARS. With the experience and expertise gained over the years, consultancy services are also provided by the Academy in its mandated areas.

NAARM is located in the premises of Acharya N.G. Ranga Agricultural University in Rajendranagar. The Academy, spread over 50 hectares area, has enviable sylvan environs. Over the years, the Academy, with support from ICAR, has been able to build up an excellent infrastructure, state-of-the-art video production laboratory, agricultural research information system (ARIS) unit and other facilities necessary for imparting training of highest quality. The Academy is endowed with excellent facilities like Library and Information Centre, Conference Hall, Seminar Hall, Committee Room, Scientist Home, Halls of Residence, Computer Lab, Offset Printing Press and Health Centre.

A striking feature of the Academy, apart from the infrastructure, is its quaint mixture of the new and old. It is this very spirit of the Academy, which is personified by the trainees and all those who work in it, trying to uphold values of integrity, earnestness, commitment and idealism. It is perhaps this spirit of the Academy that each one carries as he or she leaves its august portals to step into the National Agricultural Research System.

Mission

To enhance the performance of NARS by building capacity in research and education policy, planning and management, and to foster a scientific culture that can make the NARS highly productive globally.

Mandate

NAARM is charged to fulfil the following mandate:

- To organize and conduct training programmes in agricultural research management for the scientists at various levels
- To build up high quality resource material in agricultural research management based on actual field experience
- To undertake systematic review and study of management problems of agricultural research institutes, programmes and systems
- To plan, organize and conduct workshops and seminars in research management and educational technology

- To organize, liaise and coordinate programmes of international cooperation in the field of agricultural research management
- To assist the State Agricultural Universities in developing Regional Centres of Management to cater to the needs of various Universities

Objectives

- To organize foundation courses for the scientists newly recruited into the Agricultural Research Service of ICAR
- To design and organize training programmes, workshops, and seminars, and to develop high quality material in agricultural research and education management for the senior and middle-level scientists of NARS
- To play a major role in different processes of human resource development of NARS, in the context of agricultural research and education
- To undertake policy-level studies on prioritization of research and resource allocation, and to enhance research productivity of the constituents of NARS and their programmes
- To evolve suitable management techniques for effective planning, scheduling, organizing, monitoring and evaluation of agricultural research and development projects in NARS
- To study the management issues of various programmes and patterns of technology transfer, develop effective systems and strategies for effective communication of agricultural technologies
- To undertake development programmes for evolving more effective and efficient approaches, methods, and models in higher agricultural education for use in SAUs and Deemed Universities of ICAR
- To promote and execute academic programmes of higher learning in the specialized areas of agricultural research and education management
- To develop expertise in information technology to meet the application software needs of NARS
- To undertake systematic review and to develop appropriate systems for efficient and effective administrative and financial management in NARS
- To function as a repository of ideas and information in the form of databases in the field of agricultural research and education management, and to act as a clearing house for information dissemination through national and international networks in these areas
- To offer consultancy services in the field of agricultural research and education management, and agricultural information and communication technology.

Organization and Management

The Academy is headed by the Director. The Director is assisted by two Joint Directors and thre Division Heads in the execution and implementation of various programmes. The Academy is organize into three functional divisions, viz. Agricultural Research Systems Management and Policies, Huma Resource Development, and Information and Communication Management. The Institute Manageme Committee (IMC) guides and supports the Director by periodic review of programmes and approval investments in new areas of research and education, training programmes, workshops and seminars. T Research Advisory Committee (RAC) and Staff Research Council (SRC) provide broad guidelines a assist in developing and implementing specific research programmes and projects.

NAARM ORGANOGRAM

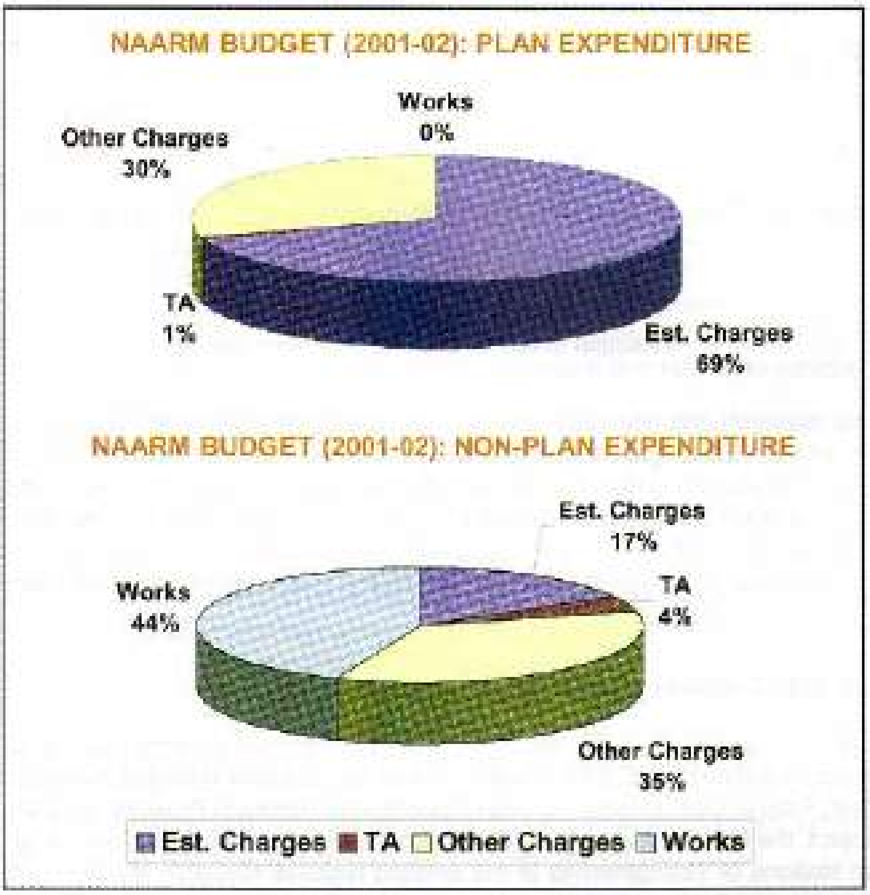
Human Resources (as on 31.03.2002)

Category	Sanctioned Strength	Posts Filled	Vacant Positions
1. Research Management Positions	03	-	03
2. Scientific	40	21	19
3. Technical	66	56	10
4. Administrative	48	45	03
5. Supporting	41	40	01
Total	198	162	36

Financial Resources (as on 31.03.2002)

The budget allocation and expenditure during 2001-02 are given in the following table. Information presented in the table below provides details on finances available from other sources and revenue generated during the reported year.

Scheme	Head	Budget allocation (Rs. in lakhs)	Expenditure (Rs. in lakhs)
Non-plan	* Establishment Charges	278.00	242.93
	* Travelling Allowances	3.00	2.96
	* Other Charges	104.00	103.96
	* Works	—	—
	Sub-Total	**385.00**	**349.85**
Plan	* Establishment Charges	30.55	30.55
	* Travelling Allowances	8.00	7.53
	* Other Charges	64.45	64.41
	* Works	80.00	80.00
	Sub-Total	183.00	**182.49**
Other Sources	* A.P.Cess	11.81	9.59
	* AHRD	73.35	55.58
	* NATP	131.17	27.20
	Sub-Total	216.33	92.37
Grand Total		784.33	624.71



Income generation

S No	Item	Amount (Rs. in lakhs)
1	Training	19.02
2	Consultancy	0.34
3	Sale of Books, Technology, etc.	0.91
4	Sale of Agri. Produces	1.12
5	a) Room rent / Licence Fee	5.75
	b) Interest on Loans and Advances	1.81
	c) Leave salary / Pension consultancy	0.13
	d) Miscellaneous Receipts	4.18
	Sub-total of a-d	**11.82**
	Sub-total of 1 - 5	33.26
6	Interest on Short Term Deposists	27.80
	Total Receipts	61.06

Distance Training in Agricultural Research Management

-A Case Study Approach to Reach the Unreached

Jagannadham Challa
Principal Scientist (Educational Technology)

Agricultural research and education have played a very crucial role in the spectacular and peaceful green revolution and shall continue to play a major role in the coming decades. The strong and vibrant National Agricultural Research System (NARS) has the primary responsibility to rise to the future challenges with its multitude of scientific and academic staff working around the length and breadth of the country. The Indian NARS with its vast network of State Agricultural Universities (SAU) and research institutes of the Indian Council of Agricultural Research (ICAR) provides the vast infrastructure and manpower for carrying out agricultural research and education.

Manpower in agricultural research

The growth and spread of ICAR has been phenomenal starting from 1966 till date. There are at present 173 research institutes in the form of National Institutes, Central Institutes, National Bureaus, National Research Centers, Project Directorates, All India Coordinated Reseach Projects (AICRP)and others. World Bank funded project, the National Agricultural Research Project (NARP), facilitated either the establishment of field research stations or strengthening of the existing regional research stations (attached to SAUs or ICAR institutes) to cater to the local agricultural research needs of the agro-climatic zones which are located in remote areas to undertake research programmes to solve the technological needs of the rural regions. At present there are several hundred research stations in 131 agro climatic zones spread across the country and more are being added as per the needs of the local farming community. Several thousand scientists and extension staff work in these stations. ICAR and SAUs are big employers in the field of agricultural education, research and extension. There are more than 30,000 personnel in different categories of scientific and technical cadres in the Indian NARS.

Competency development for research management

Competency requirement in the area of management is nowadays inherent to every profession and more so in the field of agricultural research considering the vast investments and manpower involvement. Professionalism at every level and in every sphere of agricultural research activities is now under more focus than ever before. In this context an investment during the next couple of years to develop management curriculum for agricultural research and an effective on-the job training strategy, which is cost effective and readily accessible is now a priority on the agricultural research front.

All the SAUs and ICAR institutes provide necessary technical and extension services to the farmers at the field level. In order to be functionally effective, the scientists and research managers have to develop a variety of competencies and skills and constantly update periodically. They need to learn new techniques of participatory methods that are gaining prominence in the recent years to understand farmers' perceptions, preferences and priorities more effectively. The traditional communication methods have undergone a sea change with the advent of modernization and involvement of groups of farmers. Distance education (DE) and information processing techniques using print medium, video-cum-satellite and digital based communications are in vogue. Production of self-study materials using information data banks to study and sort different appropriate technologies through easily understandable packages presented in audio-visual-animation form as extension tool kits are likely to be the new features of the paradigm shift in transfer of technology (TOT) systems. Hence, training of scientists and research managers is now a more powerful need for effective research management and technology transfer and it is a continuous requirement for agricultural development leading to overall development and alleviation of poverty. Hence, in the immediate context the need for developing research management skills among scientists working in the field research stations through alternate training strategies is paramount for NARS. This paper conceptualizes the distance training strategy for developing research management skills and the case study approach that has been initiated.

Alternative approaches of training

From 1976 to 2002 NAARM organized a series of training programmes / workshops, both at national and international levels. So far NAARM has trained 15, 276 participants through 574 programmes, with different levels of participation in areas of agricultural research management, education and management of transfer of technology. Conservatively estimating that there are approximately 27,000 scientists in the Indian NARS and assuming that at least one training opportunity for each scientist in the management specializations, the effort does not suffice the logistics requirement. Moreover the middle and senior level programmes are restricted to periods ranging from 3 days to 2 weeks, thus reducing the comprehensive coverage that could be possible in longer duration programmes. It was recommended that NAARM should develop programmes in research management for Research Management Positions (RMP), heads of research divisions, principal scientists and other middle level scientists of the entire NARS as a part of management development effort in research institutions (A. L. Choudhry Committee Report, 1995). It obviously indicates the importance that is being given to develop the competency in research management in NARS.

NAARM is training annually 350-400 scientists and academicians at present in various specialized skills in management sciences and it may not be sufficient to cover the training needs of the vast manpower in the NARS. But the scientific community that needs to be served by the Academy is so vast that at the present rate of intake of participants and the existing system of *in situ* training and also the available facilities may not suffice the purpose. Alternative HRD strategies for training component have to be explored to realistically match the needs. Moreover, *the Quinquennial Review Team(QRT) of NAARM (1987-94) observed "…. considering the 'knowledge expansion' taking place in modern times, it is necessary that the scientists already in service are provided at least three in-service training opportunities during their career. This means that in this regard the training load would be something like 90,000 persons over a period of 25 years. According to this estimation about 3,600 persons are to be provided in-service training opportunities every year".*

It is indeed a gigantic task to accomplish and hence, a new strategy has to be evolved to explore alternative modes of training and educational programmes to impart research management and related skills. NAARM did initiate such an effort in the area of educational technology for teachers of SAUs through distance education programmes (Outreach Programme in Educational Technology, 1991-1995) on a modest scale. Similarly, the need of the organization was well recognized in that the academy has proposed to undertake distance education programmes in agricultural research management on a regular basis from the 10th five-year plan onwards (VISION - 2020, NAARM, 1998). Distance education and training strategies are increasingly being explored and have become the necessary tools in the present scenario. However, the greatly felt need now is to exploit distance training mode to provide the required management skills for the researchers and the research managers in the field research stations with a view to enhance management skills and enlarge the scope of training programmes to reach the un-reached in the NARS.

ICAR, being the apex body in the NARS, is undertaking a major modernization programmes through information and communication technology initiatives for better access to information and exchange for research programme planning and to modernize the management information systems for management of research and developmental projects. In addition, it is also aiming for improving the management competencies and practices through several organizational and management reforms and is also about to implement project based budgeting and management at micro level. NAARM is entrusted with the central responsibility to meet the challenging needs of NARS in this endeavour of imparting training and to develop new delivery strategies for competency development through in-service training programmes of the vast scientific body in agricultural research. Distance education (DE) is one such alternative whose appropriateness is now being considered seriously to offset the training costs and to provide on-station training to save time so that periodical refresher training can be more easily accessed to the scientists and research managers (Anonymous, 1998). Moreover, it is also intended to incorporate the developments in information and communication technologies so as to reach a large number of target groups of scientists working in remote research stations frequently and effectively. At this juncture it is appropriate to review the historical perspective of DE efforts of the past and to foresee the future strategies.

Distance education: Concepts and definitions

From its emergence distance education authors have referred to DE using different terminology: correspondence study; teaching by mail; teaching in writing; distance teaching or teaching at a distance; home study; independent study; open learning, external study and finally self-study (Edstrom, 1970; Keegan, 1980). Many reasons have been put forward to explain this lack of unanimity concerning the choice of terminology with reference to this non-traditional mode of education. The rapid expansion of DE and the evolution of the concept itself explain in part this variety. But, we believe that it may also reflect the lack of consensus between scholars and practitioners of DE in their understanding of the concept.

Indeed, such obvious divergence among the DE professionals is translated in the definitions that they use to describe the concept. Until the 1970's, the educators in the field were less interested by the study of the concept and by theoretical considerations. They were mostly devoting their energies to the logistics of the DE enterprise, and the literature was generally descriptive of those efforts (Brown and Brown, 1994). This lack of theoretical background has certainly contributed to the rejection of DE by many educators belonging to the traditional mode of education, and by decision-makers in education (Edstrom, 1970; Keegan, 1980; Perraton, 1982).

From the 1970's onward, we have witnessed more conceptual work. Keegan (1983) classified the major theoretical contributions under three categories, reflecting the different emphasis of various scholars:

- Autonomy and independence;
- Industrialization;
- Interaction and communication.

Michael G. Moore (U.K.), a leading spokesman of the "autonomy and independence" category, stressed that the most important dimension in the concept of DE is the distance. He defines distance, not in the geographical sense, but in terms of the distance that is created by a given DE activity. He proposes to measure the distance through two variables. The first one is the type of dialogue created by the choice of media and their utilization (design). Indeed the variable dialogue, which is a measure of the degree of the two-way communication built in the DE training-learning couple, varies with the type and use of the chosen media. The second variable affecting distance is the structure, which stands for the measure of adaptability or flexibility of the DE programme. For Moore (1983), a program where all materials are prefabricated and all the steps in the training process are predetermined is most distant because it is not sensitive to the self-defined needs of the learner and not responsive to his/her particular set of circumstances. This concept of DE focusing on learner's training aspirations and environmental constraints is of particular significance in dealing with adult learners in distance training (DT) situations.

For this scholar, another important dimension is the learner-autonomy that he defines as "the extent to which the learner in an educational programme is able to determine the selection of objectives, resources and procedures, and the evaluation design" (Moore, 1983). This raises the question of the type of learner who would benefit most from a more distance or less distance programme. For Moore it depends on the capacity of autonomy of the learner, and since this capacity varies from one type of learner to the other, it becomes crucial that the choice of media, the design and the structure of any DE programme correspond to the type of target learners. One last important aspect to look at is that the DE programme should allow each learner to express his capacity of self-autonomy, and to grow through his learning process. It is worth noting that for this author, the role of the educator is crucial as he is the one who should be helping the learner to rediscover his/her own capacity of autonomy, which may have been inhibited or destroyed by a previous traditional formal education. (Moore, 1983).

The contribution of Otto Peters of Germany, labeled "theory of industrialization" represents the second category of concepts on DE. It borrows some principles from the industrial world to explain the didactic structure of DE. Opposed to other scholars, who emphasize the individualistic aspect of DE, he suggested that we should conceive DE as one kind of prefabricated teaching. According to him, this characteristic distinguishes DE from the traditional face-to-face mode of education, since this type of education does not anymore have to rely on a personal communication, but rather on an industrialized and technological one (Keegan, 1980). Peters 's concept of DE is still very controversial because it neglects what really matters in any form of education. Even though we may not agree with his theoretical vision, we have to recognize that his contribution has the merit of drawing attention on the industrial characteristic of many DE programmes, and on the unfortunate consequences of it on the life of the teacher, not to mention the learner.

The third category, regrouping theories of *"interaction and communication"* is often associated to Borje Holmberg of Sweden. For this author, the essential characteristic of DE is "non contiguous communication" (Holmberg, 1985). He identifies two descriptive elements related to this type of communication: the pre-produced nature of the learning material, basis of the distance study, and the design of the two-way communication. While acknowledging the autonomy of the learner and that learning is an individual process, Holmberg does reject the notion of mass production of distance teaching as described by Peters. *The term "self-study", also often associated with this scholar, implies here that the learner has the capacity to assume the responsibility for his learning.*

From this brief overview of some concepts of DE, it appears that, in the field of DE, two schools of thoughts clash. At one extreme, we find the industrial type of DE that emphasizes individual study with the use of centrally produced learning material. This is considered the most cost effective one since it relies on economies of scale in production and on distance tutoring which comes at a lower cost than formal teaching. This type is also perceived as totally different than the traditional face-to-face mode of education. The classical example is the correspondence course that has been pioneered by the Open University in the UK (and developed by WEP in specialized fields). At the other extreme, there is the "small scale" type of DE that considers the face-to-face element as a regular component of DE activities. It is extremely important to stress that the face-to-face element does not have to take the form of a workshop, but may take the form of a support group (Holmberg, 1985).

Between these two extremes, there are more middle views that proposed that the *face-to-face* element should in fact vary with the content to teach at a distance. Unfortunately the distance educator often does not trust enough the communication medium potential, or does not know how to use it in a creative way. Perraton (1981) believes that the dialogue element should be considered as an essential component of DE, if we want to speak of education, and not indoctrination. In order to ensure the presence of the dialogue element in the distance educational process, he encourages the use of other means than the typical face-to-face element like the workshop, as for example the support group or discussion group. According to this scholar, since the dialogue element is the most essential component of any DE activity, it should be the one to which the distance educator should pay the most attention.

This short review of the literature on some of the concepts of DE suggests that there are many ways to define DE, each translating the scholar's vision of the concept. Some authors even say that "DE is more an "encompassing descriptive label" than it is a true definition" (Knott, 1994.). Nevertheless, we have decided to focus on two definitions. The first one is presented here because it is one used by professionals in the field of DE who are mostly interested by distance adult education, as it is the case in the project on Distance Training for Agriculture Research Management[3]. So in the context of adult education, Henri and Kaye (1985) defined DE as:

"...a form of assisted autodidaxy which allows the adult learner to access mediatized resources of knowledge without the classical intervention of a teacher but with the support of a network of resources".

This definition is precious since it reveals how distance, in the DE concept, modifies the traditional responsibilities attributed to the teacher/trainer and the learner. Moreover, this element introduces a *third partner*, in the "andragogical relationship", whose presence and contribution become necessary in the unfolding of the educational activity: the "intermediary" person, labeled "tutor", "facilitator", or "counselor". At Indira Gandhi National Open University (IGNOU), the later term is favored. The presence of a third partner certainly changes the dynamic of the educational relationship. In short, the distance element, because it imposes the use of some form of communication, and requires new roles to be played by the partners of the DE triad, creates an educational relationship not only different, but much more complex. The weakness of this definition is that it seems to leave minimal space to the educational organization, giving the impression of neglecting the role any institution should play in any DE activity.

It is clear that this definition proposes a very idealistic portrait of the adult learner. As mentioned above, more than often, the reality in DE is that the target adult learner has lost, or never acquired, the essential characteristic upon which this definition and its concept of DE relies: learner-autonomy. However, we consider that this theoretical contribution is not to be left aside, since it certainly opens new avenues of research in the field of DE. Keeping in mind the richness of this definition, we propose to look at a second definition, chosen for its simplicity and for its complete openness. So, for the purpose of this paper, and our future research work, we will propose that DE should be defined as:

"... any sort of instructional activity during which the learner and instructor are physically separated for part or most of the time" (Knott, 1994).

In other words, DE takes place when a teacher (trainer) and student(s) are separated by physical distance, and technology (i.e., audio, video, data, and print) is used to bridge the instructional gap (Willis, 1994). As the above definition indicates and, as it was underlined in above discussion, DE can take different shapes, from a very simple one to a very elaborated one. Besides the fact that its design and structure may differ, DE can be brought in for many different purposes. As Knott (1994) mentioned, with the ongoing development of the telecommunication technologies, the uses for DE are even more varied, making this non-conventional educational mode even more attractive. But the potential of DE does not simplify the task of teaching/training; quite to the contrary it is complicating even more the life of the professionals in the field of DE. In this paper, Distance training (DT) will be used to refer to the organizational framework and process of imparting agriculture research management skills at a distance. However, the intended instructional outcome, or the learning that will take place at a distance, will be referred to as Distance learning (DL).

Distance education in the Indian context

Since independence, distance education systems have existed in India for school dropouts to complete school leaving certificate so as to pursue higher studies in social sciences. Later, home study programmes were offered by some universities for bachelor's degree and subsequently even for master's programmes. Gradually several professional associations used this system to offer diploma programmes in Chartered

Accountancy and Engineering Sciences. Delhi University was the first to establish a Directorate of Correspondence Schools in 1962 and several other universities followed the trend. Thus, distance education in India was firmly established in the university system. At present there are about 50 institutes offering correspondence courses. The need for institutionalization of DE was recognized and the first open university in India was established in 1981 in Hyderabad, Andhra Pradesh Open University which was later renamed as Dr B.R. Ambedkar Open University (BRAOU). It offers Bachelor's degree in arts, science, commerce and library/information science. To its credit it established 57 district study centers offering instructions through print, audio cassettes, radio broadcasts, face-to-face tutoring and counseling.

A few years later, came the need for a National Open University to offer quality education to a much larger population that remains outside of the university system. Consequent to the passing of an Act in the Indian Parliament, Indira Gandhi National Open University (IGNOU) was established in 1985. Very soon IGNOU started programmes such as the Diploma in Management. IGNOU is spread around the country having 16 Regional Centers and 244 Study Centers where counseling and other student support services are provided (IGNOU, 1998). Both BRAOU and IGNOU have also opened centers in jails and other reform centers. At present IGNOU offers a variety of Diploma and Masters programmes in several management disciplines. The University Grants Commission of India (UGC) regularly sponsors video based educational programmes through Electronic Media Research Center (EMRC) on the national TV network.

The Maharashtra Open University (YCMOU) offers programmes in agriculture, teacher education and communications. One professional association called HRD Network of India offers postgraduate diploma programmes in Human Resources Management through distance education mode. Recently, the Administrative Staff College of India (ASCI), an institution in the private sector for business management has entered into collaboration with British Universities for offering distance training in business management through electronic media for the executives of the corporate world.

More and more programmes are being explored to be offered by distance education mode. Essentially, distance education is now considered as the only option to save the constraints of time, distance and finances, to cater to a larger audience and to provide opportunities for learning which otherwise would not be available. The programmes offered by the two open universities in the country are quite popular and the cafeteria of programmes is being increased as more and more areas and skills are being brought under the ambit of distance education. It was envisaged that by the year 2002, computer net-work of the open universities and correspondence course institutes along with the regional and study centers will establish Open Education Network (OENET) for catering to educational needs of the millions. The National Education Policy (1986) observed that the future thrust will be in the direction of open and distance education (MHRD, 1986). It can be concluded that distance education is currently in the stage of evolution to assess the present state of this system and it is going through the developing stage. For example, distance education is still suffering from such attributes as second choice of learners, lack of recognition of the diplomas for jobs, high cost of study, lack of self-learning and interactive study materials, and lack of awareness etc. At present the predominant medium used for distance education in the formal academic programmes is still print medium, video and video-conferencing have been explored and are expanding.

Specificity of training needs in agricultural research management

Distance education in the field of management skills has by and large remained in the formal academic programmes so far through the open-university system or through the professional bodies. A variety of courses are offered mostly catering to the needs of the business sector leading to award of degrees or diplomas. Distance training as a training mode for on-the-job training is still a novelty in this country. Distance education as a training-package for in-service training is being considered by ASCI for business executives. The flexibility-of in-service distance training mode is an important feature. The exploring of such methods for imparting management skills through distance mode does indicate that it is possible to impart management training while the executives are on their day-to-day responsibilities (Leonard, 1996). Distance training also appears to be more effective since the learner can relate his experiences while interacting with the self-learning study materials at his own pace while being actively on the job.

Training in management skills involves study material, case studies, illustrations and examples, simulations and games, group discussions and assignments. In distance training mode the interactive lecturing and group discussions need to be substituted by interactive study material in a discussion mode while all other forms of learning can be retained even in distance training approach. Management skills in agricultural research are more appropriate on the job while the scientist and the research manager are actually experiencing the need for the management skills rather than in a classroom situation. Quite often the feedback has been that the management training capsules offered in the institutes are far from reality and mostly academic in nature and of scholarly value. In this process the distance training approach can offer "experiential learning" advantage. Research management skills need to be constantly updated with the changing research priorities, technology availability and one time training programme may not be sufficient as the nature of responsibilities of scientists are constantly changing. Thus distance training approach offers an immense potential for imparting management skills in agricultural research and reaching a large clientele group in the shortest possible time.

Identification of strategic options for distance learning for NARS

It is worthwhile to identify possible strategic options for the preparation and delivery of DL/DT for agricultural research management in NARS. The choice of option has to be made on the basis of three inter-related themes concerned in turn with:

- The effective transfer of management skills and knowledge in agricultural research management using a DL system;
- The need to maximize the interactive nature of the DL materials produced;
- The actual preparation and presentation of the DL materials.

Clearly with the rapid development of new technologies that are effectively revolutionizing the production and delivery of learning materials for DE/DT, it was important to consider a range of options that took this into account to varying degrees. There are three broad models:

- A "technology input" model that envisaged the rapid incorporation of computer-mediated and computer-assisted learning, aligned with a predominantly electronic delivery system;
- A "mix and match" model that saw a progressive adaptation of learning materials from essentially print-based to multi-media with an increasing emphasis on computer-based learning.
- A "low technology input" model that emphasized a continuing reliance on print-based delivery in the medium term at least, with some developments into multi-media, but fell short of an over dependence on the latest developments in ICT.

To decide on the selection of a preferred strategic option, the following key issues were focused, which in effect provides a reliable checklist for similar undertakings that are concerned with capacity building using a distance learning approach:

- Medium of learning delivery
- Quality of learning materials
- Cost of preparation and implementation
- Accessibility of learning materials

- Flexibility in relation to learning activities
- Infrastructure availability for delivery and support
- Suitability of the learning materials to encourage self-learning
- Appropriate instructional design
- Learner support system
- Importance of partnerships and collaboration for sustaining delivery
- Role of NAARM resource persons and on-site facilitators
- Quality assurance in terms of writing, editing, packaging and delivery
- Possibilities for franchising the learning materials
- Evaluation of learning outcomes

A preferred strategy for distance training for NARS

Evolving from a lengthy and fruitful consideration of the issues outlined above a concise statement of the preferred strategy and a series of related themes will provide the core of the strategic approach envisaged at NAARM and the potential limitations and constraints there of.

It is worth pointing out that while there is a general acceptance of the need to take due account of the developments in information and communication technology (ICT) in the choice of the preferred strategy, it was also decided that technology should not be seen to drive the educational process. Rather the educational process should work towards the progressive integration of new techniques of preparation, presentation and learning as appropriate. The question of quality assurance is always seen as uppermost in this endeavour.

The preferred strategy has the following premises:

- Optimizes the use of existing resources and facilities at NAARM and at NARS;
- Encourages collaboration and partnership in implementation;
- Builds upon the essential theme of participation, which includes teamwork and multi disciplinary team building;
- Envisages a phased use and integration of distance learning media, from print-based with audio-visual supplementation to the increasing use of ICT developments.

The following paragraphs provide a clear indication of the direction in which NAARM will need to travel to ensure effective development of a system of DT for agricultural research management that is sustainable.

Learner profile: The needs of the learner, in effect one of the "stakeholders", must be explicit and to the fore. It is envisaged that the primary group of learners will be ICAR and SAU scientists and managers at head of division level and below who will be seeking to improve management skills. Clearly, it will be beneficial if the potential users of DT/ DL facilities are already performing some type of management function. On-the-job education and training will certainly require employer support and a time allocation from working hours. Actual learning may be on individual or group basis, depending upon the topic and nature of the training programme and target audience.

Content of training programmes: Content will vary according to the target group being served. It should always be needs driven, learner based and learner defined. In general terms, the subject matter will be concerned with managing research, managing people and managing the resources. It will be based on actual and evolving management processes in the ICAR and SAUs. The actual process of capacity building should be a gradual one, broadly following this formulaic approach:

Awareness ⟶ Knowledge ⟶ Skills

Planning and management: The primary concern is to provide a research-based, non-formal, in-service opportunity for quality training in agricultural research management, to "reach the unreached" scientists, particularly those who are located in remote research stations in rural areas. For optimal planning and management there will be need to build close liaison and collaboration with all the institutions of NARS. This has to take into consideration the administrative and financial implications for the learners, time to be spent on the training, the cropping seasons and access to the "focal points of learning or learning sites".

Implementing quality assurance: Quality checks need to be built in at successive stages of the course development process, including material production. Writers of resource material must be based on merit and experience and there should be frequent feedback from reference groups. There is need for independent professional editing and reviewing both in terms of content and DL approach and style.

Cost effectiveness: There will be costs to both NAARM and learners. At the institutional level there will be significant infrastructural and operational costs to be borne. A major investment will be required for the progressive build-up of technology and improved access generally to communication facilities. A sizeable slice of the budget will also go towards development, production and packaging of learning materials. There will also be a need for budget for communication, to enable NAARM facilitators to travel and for learning material distribution, monitoring and evaluation. For the learner there will be a cost to bear for the materials, for access to computers and internet as required and for transport to face-to-face sessions at the learning sites. The more meaningful the learning materials and the more focused the training, the greater are the likely costs to be incurred. There is clearly a need for sound cost-benefit analysis.

Use and integration of media: As clearly stated in the statement of the preferred strategy above, it is envisaged that the print medium will be used initially, progressively supplemented in phased development stages over a period of three to five years by interactive audio-visual and computer mediated learning.

Instructional design and production: It is clear that instructional design and production will be in distance mode with face-to-face sessions among the learners at the selected learning sites. Course materials will be customized as far as possible for the target audience, and there will be considerable emphasis on the use of case studies and illustrations. Selected texts on management will also be supplied. Of paramount importance is the need to develop a "ready responsiveness" and user-friendly access in the preparation and presentation of learning materials. To ensure a participative approach, the learning resources should incorporate diagnostic grids related to specific training topics and will be required to complete periodic response sheets.

Learner support system: A sound and effective learning support system is very important for the success of the DT in agricultural research management. Essential provisions include institutional support in hardware, reliable feedback mechanisms, well trained facilitators and local coordinators, administrative clearances and financial allowances to learners, recognition of DT for career advancement, effective communication channels for interaction.

Measuring learning outcomes: In many DT/DL programmes, the assessment and measurement of learning outcomes is often the most neglected and least developed element, yet it is arguably the most significant. Critically there must be a prescribed and explicit connection between criteria established for admission to the DT programme and the outcomes, both expected and ultimately achieved. This requires a continual process of monitoring and evaluation, revision, adaptation and update. A suitable assessment mechanism should be developed with respect to accepted and agreed principles of adult learning. The importance of self-assessment cannot be underestimated both as a pre-training activity to form the basis of "learning contract" and to provide evaluation variables for the appraisal of training programmes.

Partnerships: Last but not the least is the issue of partnerships. To succeed in capacity building project like this requires cooperation and collaboration. No single institution can adopt a "go-it-alone" philosophy. Ideally, the concept of partnership would be realized here through the development of a consortium of NARS institutions, with NAARM identified as the lead agency. Potential collaborators/partners could include SAUs, IARC, IGNOU, IIMA, ASCI and other management and rural development institutions likely to have an interest in this venture and its anticipated outcomes, linked to achieve sustainable rural livelihoods and the alleviation of poverty.

Possible constraints and limitations

The preceding paragraphs provide considerable substance to the initial vision, albeit in brief form. This is an exciting prospect and yet every challenge has its drawbacks. There has to be a system of checks and balances and there can be no ignoring the potential constraints and limitations. This list below is neither exclusive nor wholly inclusive.

- Target audience will be voluntary, comprising both scientists and all other management functionaries;
- Lack of incentive and/or motivation for the learner
- Lack of recognition of the DT progarmme credentials at national level by ICAR
- Isolation of the learners: group learning opportunities may be limited and may vary considerably from one region to another
- Underestimation of what is required in resource terms to produce cost-effective quality learning materials
- Developing an adequate infrastructure for learning
- Financing the training programmes
- Marketing the training programmes
- Possible lack of support from potential partners
- Shifting of learners from their place of work
- Overload of work during cropping seasons

Preferred distance training strategy to distance training package

The strategy that was designed above was to provide a basic model or framework to be tested in a pilot programme to start with. A pilot distance training programme (PDTP) was developed by the project to be tested in five SAUs viz. ANGRAU, Hyderabad (Andhra Pradesh); MPKV, Rahuri (Maharashtra); UAS, Dharwad (Karnataka); TNAU, Coimbatore (Tamil Nadu) and TANUVAS, Chennai (Tamil Nadu). After a series of brainstorming sessions the following issues have been considered for developing the pilot distance training module and its implementation.

- The entire approach will be for developing the training package in a practical mode.
- There will be no formal or academic learning in the training.
- The learners of the PDTP will be put through a series of practical exercises and assignments to bring them closer to villages, using sound adult-learning principles.
- The interactive nature of PDTP calls for more emphasis on the practical exercises.
- More attention will be given to collaborative learning, team building, interpersonal skills, communication skills with farmers and such other skills as writing, presentation, etc.
- Evaluation mechanism will include both formative and summative evaluation.
- Linking of agricultural research with poverty issues is mandatory and hence efforts are made to educate scientists on poverty issues.
- 14 learning sites were selected in five SAUs for implementation of the programme.
- Each learning site facilitates a group of five learners who are mostly drawn from the research stations around the learning site at an average distance of two hours travel by road.
- The pilot programme will be tested on a total of 70 learners.

In the end, the following two major objectives emerged for the PDTP:

- Focusing agricultural research on poverty alleviation
- Developing managerial skills

The module was titled "Focusing Agricultural Research on Poverty Alleviation" and structured as a 16-week course with the following learning objectives and expected outputs.

a. Learning objectives:

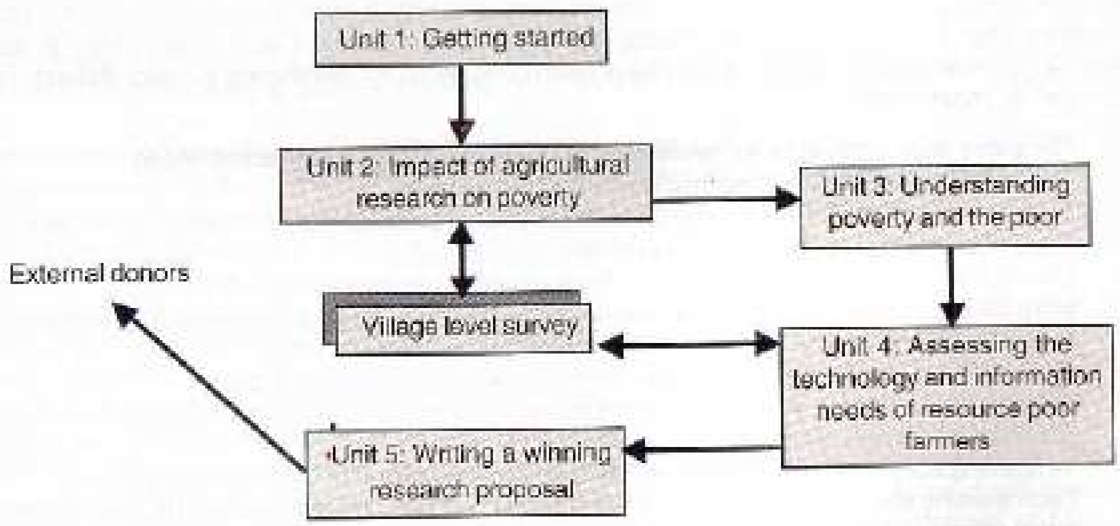
- Increase the relevance of agricultural research projects in the future to the needs of the poor.
- Improve the standard of research proposals, again focused on poverty reduction among resource-poor farmers.
- Provide the opportunity for developing practical skills in group-work and village-level enquiry.
- Create a body of empirical results from experiential 'action-learning' that can be used for future agricultural research at the research station and agricultural university level.

b. Expected learning outputs:

- How to write successful research proposals with a focus on poverty alleviation.
- How to identify and describe resource-poor farmers (RPF) and characterize their needs through village surveys.
- Transferable skills derived from group-work and action learning. These include interpersonal skills with clients and fellow scientists, communications skills, team work, problem solving and presentation skills.

c. Course structure:

The diagram below sets out the main elements in the course.



Expected outputs from the course:

The following outputs are expected from the learner groups:

- Seventy trained scientists in "focusing agricultural research on poverty alleviation".
- Fourteen village level survey reports on the impact of agricultural technologies on resource poor farmers.
- Fourteen village level monographs (one per group of five learners). These will give insights into (a) the characteristics of poverty in the villages and (b) an assessment of the technological and information needs to overcome poverty. The output obtained by teams of learners who perform well may also be used to publish a paper on the subject.
- Fourteen research proposals focusing on poverty alleviation (one per team of five learners).

d. Study calendar and study time

It is expected that each participant should devote about 8 hours per week and 150 hours in total to the course. Most of this time will be spent working in a group with fellow participants rather than individual study. The course is divided into five units of approximately 35 hours each. The Unit 1 is a small unit for getting started. Each unit will include some reading, practical exercises and fieldwork where practical experience would be gained on the topics of the unit. Each unit is further divided into sub-units corresponding to one week of work each. The pilot test of the module will be conducted over a five-month period in 2001-02, timed to coincide with the *Rabi* cropping season.

Study calendar	Topic	Week	Self learning	Group learning	Village survey
Unit 1	**Getting started**	1	✓	✓	
Unit 2	**Assessing the impact of agricultural research on resource poor farmers**				
2.1	Establishing an overview of technologies released	2	✓		
2.2	Choosing a methodology for impact assessment	3	✓	✓	
2.3	Impact assessment of technology on resource poor farmers-village visit	4		✓	✓
2.4	Lessons learnt from impact assessment of agricultural research on resource poor farmers	5	✓		
Unit 3	**Understanding poverty and the poor**				
3.1	Understanding local perceptions of poverty and the poor	6	✓		
3.2	Poverty alleviation and agricultural research	7	✓		
Unit 4	**Assessing the technology and information needs of resource poor farmers**				
4.1	Review of approaches to identify technological needs of RPF	8	✓		

4.2	Participatory Rapid Rural Appraisal: Review of methods	9		✔	
4.3	Field visit to the village and application of PRA	10		✔	✔
4.4	Writing the technology and information needs assessment report	11		✔	
Unit 5	**Writing a winning research proposal**				
5.1	Writing a concept note	12	✔		
5.2	Selecting and justifying a group research project	13		✔	
5.3	Drafting the full research proposal	14	✔		
5.4	Peer review of research proposal	15		✔	
5.5	**Course evaluation and conclusion**	16		✔	

e. Training strategy:

The training strategy adopted has several aspects:

- **Structure:** The course is divided into five units of roughly 35 hours each in four units with one small unit to get started at the beginning.
- **Assessment:** There are no examinations and participants are guided in ways to assess their own performance during and at the end of the course. Also each team must submit a report showing feedback on each unit and a research proposal based on their fieldwork of Unit 4 at the end of the course.
- **Facilitation:** Administrative and other possible help will be available from the Site Coordinators at the learning site chosen for hosting each team. Site Coordinators are trained to facilitate the learners in their learning activities of the course without interfering or imposing.
- **Learning process:** Distance learning with face-to-face components among individuals and an emphasis on *'Self-learning'* through experiential, interactive, collaborative and peer learning.
- **Monitoring:** NAARM team will carry out periodic monitoring and evaluation during and at the end of the course. The designated NAARM resource person will visit each learning site for a minimum of three times during the course.
- **Training kit:** Course pack (working manual in print medium) and a practical kit for a simulated group exercise.

This training programme is designed in an action-research mode wherein the participants are both *"the observer and the observed"*. We share what we have and invite participants to join us in an effort to expand the knowledge on the subject of "**Focusing agricultural research on poverty alleviation**". At the same time the trainees go through an experiential training-learning process to acquire appropriate managerial skills. This is a basic premise of this "distance training" programme, i.e. to combine research and training in a case study approach.

Justification of the training strategy

In a distance-training mode, face-to-face sessions may be fewer and/or less direct and we substitute them with communication technologies such as self-instructional printed materials, correspondence, written comments provided on assignments, email and telephone mediated interaction. Thus, the resource persons from NAARM and trainees perform their essential tasks and responsibilities of training-learning far apart from one another, communicating in a variety of ways. This system encourages the group of learners to be autonomous and to develop a capacity to carry out self-directed learning to acquire skills.

The proposed program will attempt to turn the isolation of the average distance learner into an advantage and use his work environment to the utmost to generate new knowledge while acquiring the training. Why role-play in a classroom theoretically an encounter with a poor farmer when it can be done practically in an actual real situation? The advantage of the classroom simulation is to provide some feedback to the trainee on how he/she did. This can be substituted by having trainees themselves observe the situation and report on what they saw, and receive feedback on their reports.

This requires that training be set in a group process. This is the second premise of the approach. In this form of group learning in each practical exercise, one trainee is assigned the role of observer. The observer reports to the group what he observed. This forms the basis for the debriefing after each exercise and also the basis for the comments by the site coordinator. Each trainee in turn acts as an observer. A format for recording the observations of each group session is provided in the training manual at appropriate places.

The resource persons from NAARM consisting of one Principal Investigator and three team members provide structure to the learning programme by periodic monitoring and evaluation through site visits and participate in occasional sessions with trainees in analyzing the outcomes and record the progress of the training-learning process. They are also available on-line either on e-mail or telephone/fax for interacting to seek clarifications, guidance and answers to specific questions from the trainees and the site coordinators.

The distinction between trainers and trainees (or learners) becomes quite meaningless in this context since NAARM resource persons and site coordinators will also be learning along with the learners. This approach could be applied to most aspects of research management where the knowledge available may be inappropriate for a rapidly changing environment. This suggests that the approach used in this pilot project could become a new approach not only to train scientists but also to generate new knowledge about the management of research, the people, the programmes and the organizations. Hence, the term "action-research" mode is justified.

1. Learning strategy:

The recommended learning strategy for the learners consists of three phases. First, the learners work through the topics in the sequence in which they appear in the study manual. They do this as a team. Second, in each unit, the learners discuss together the issues and ideas as a group. Third, they complete the practical exercises and reflect on the outcomes as a group. The recommended learning strategy envisages the following four distinct types of learning apart from self learning activity.

- **Experiential learning**: There is a fair amount of text written for this course, which is mostly informative. Participants taking the course are encouraged to spend more time acquiring practical skills through group discussions and field exercises.
- **Interactive learning**: Although face-to-face contact sessions with resource persons are not structured in the programme, the participants are encouraged to interact with the resource persons of NAARM through e-mail, telephone/fax to seek clarifications, guidance and answers to specific questions during the course of the programme.
- **Collaborative learning**: Individuals learning through group processes is the third feature of the strategy. Team-building and group dynamics are in-built at the beginning of the course (unit 1). These skills are then required for the to complete the rest of the course. Collaborative learning is group-based aiming at development of following skills.

- Interpersonal skills with clients and fellow scientists
- Communications skills
- Team work
- Problem solving
- Writing and presentation skills

- **Peer Learning:** Groups learning on their sites are encouraged to interact with the other similar groups in their university or in other participating universities through e-mail, telephone/fax and/ or personal contact to share and exchange information, progress of work and potential learning. This will facilitate dynamic feedback mechanism among the groups of participants on the training-learning process.

g. Learner obligations:

- Participants are expected not to avail long leave, any other training programmes, transfers or deputations during the period of training.
- All the participants are expected to remain on duty and participate in all the learning activities without fail.
- The participants are expected to have their personal e-mail address and use it for communicating with NAARM.
- Evaluation of performance of the administered exercises and response sheets for each module will be carried out regularly.
- The time schedule of all the activities should be maintained by prior agreement in group meetings. The individual assignments as decided in the group should be carried out without fail and as per schedule
- The participants are advised to adjust their personal agenda in congruence of the learning activities of the programme and the decisions of the group
- Participants as a group interact with NAARM project team during each site visit to discuss the progress of learning and sharing of experiences and give feedback
- All the exercises and assignments as scheduled should be completed and shown to the site coordinators who in turn will pass them to NAARM
- Each group will present a group seminar on the outcome of the entire training through a research proposal to the scientists of the site/university at the end of the training programme.

The pilot distance training programme is currently in progress and a full report on this case study approach would be available in due course.

Conclusion

Distance education undoubtedly offers tremendous potential. New technologies are there, and if used with creativity, they may lead us to a very powerful mode of in-service training of a large number of scientists and research managers. It also provides the potential for continuous learning so that the need for refresher courses on a regular frequency can be achieved. NAARM has attained the capacity for imparting training in research management as different from business management and has developed study material for classroom training. But, it is becoming evident that the sheer logistics does not permit the effectiveness of the existing delivery systems to reach the desired goals. Thus, distance training strategy has to be seriously considered by ICAR-NAARM as a means to break down the barriers for researchers and research managers to access training. The research project on "Distance Training for Agricultural Research Management" proposes to explore the feasibility of such a strategy which, if proved successful in imparting management skills to remotely located scientists, may lead to the enablement and empowerment of the ultimate beneficiaries of the research: the farmers and their families. However, like any educational endeavour, if it is poorly planned and evaluated, it may contribute to widen the gap between the incredible amount of available knowledge and the presently "un-reached" scientists who need the information and the commensurate skills the most. The flexibility of DT methodology offers tremendous opportunity for adapting to various needs of NARS through pilot study schemes and to redesign the DT strategies for sustaining on long term basis for a variety of training interventions and to generate new knowledge and experience through this bold and innovative experiment.

References

Anonymous, (1998). Distance Training, http://www.weber.edu/ist/itfm/disttrng.htm

Brown, F.B. and Brown, Y. (1994). Distance Education Around the World, in B. Willis (Ed.): Distance Education, Strategies and Tools (3-35). Englewood Cliffs, New Jersey: Educational Technology Publications.

Choudhry, A.L. (1995) Committee Report on Restructuring of Training Programmes of NAARM, ICAR, New Delhi.

Edstrom, L.O.(1970). What is Correspondence Education? In L.O. Edstrom et al. (Eds.): Mass Education: Studies in Adult Education and Teaching by Correspondence in Some Developing Countries (102-122). Uppsala: The Dag Hammarskjold Foundation.

Henri, F., Kaye, A.(1985). Enseignement à distance - Apprentissage autonome? In F. Henri and A. Kaye (Eds.): Le savoir à domicile (Chap. IV, 99-144). Québec: Presses de l'Université du Québec.

Holmberg, B.(1985). Status & Trends of Distance Education (2e Ed. Rev.). Lund, Sweden: Lector.

IGNOU, (1998) Indira Gandhi National Open University: A Profile, New Delhi.

Keegan, D. J.(1980). On Defining Distance Education. Distance Education, 1, 13-36.

Keegan, D.J.(1983). Theories of Distance Education. Introduction, in D. Stewart et al., (Eds): Distance Education: International Perspectives (63-67). New York: St Martin Press.

Knott, T. (1994) Planning and Evaluating Distance Education: A Guide to Collaboration. Memphis Tennesse: Diaphera Publications

Leonard, B.,(1996). Distance Training: Work and Training Overlap. HR Magazine, Microsoft Internet Explorer, pp1-6.

MHRD.,(1986) National Policy on Education, New Delhi, Ministry of Human Resources Development, Government of India.

Moore, M.(1983). On a Theory of Independent Study, in D. Stewart et al., (Eds): Distance Education: International Perspectives (68-94). New York: St Martin Press.

Perraton, H.(1981). A Theory for Distance Education. Prospects: 11.

Perraton, H.(1982). Alternative Routes to Formal Education. Baltimore: The Johns Hopkins University Press.

QRT, (1987-1994) Report of the 3rd Quinquennial Review Team on NAARM, 1995, ICAR.

VISION 2020, (1998) NAARM Perspective Plan, NAARM, Hyderabad-500 030

Willis, B., (1994). Distance Education Strategies and Tools, Englewood Cliffs: Education Technology Publications.

Training Achievements

Imparting training in the broad area of agricultural research and education management is the key activity performed by NAARM. A brief outline of the various programmes run at the Academy is as follows.

Foundation courses: These programmes aim at improving the knowledge and skills of scientists and administrative staff at the entry level when they are recruited to the ICAR system. Programmes for the scientists focus on agricultural research management, whereas that for administrative and finance officers deal with administrative and finance management.

Senior-level / refresher courses: For the benefit of the middle-level scientists and faculty members of ICAR and SAU systems, advanced programmes are conducted on various aspects related to agricultural research management, human resource management, and information and communication management. Some of these programmes are recognized by ICAR for the Career Advancement Scheme of scientists and faculty members.

Management development programmes (MDP): These are designed as per the managerial needs of the newly recruited Heads of Division, Project Coordinators and Zonal Coordinators.

Executive development programmes (EDP): Newly recruited Directors, Assistant Directors General, Joint Directors of national institutions, who comprise the research management position cadre of ICAR, participate in these programmes.

Sponsored programmes: These packaged programmes are designed as per the needs and requirements of clients from both India and other developing countries.

Distance training programmes: To reach a wider audience, distance training programmes in Agricultural Research Management and Educational Technology are offered for developing the capacity of scientists and faculty members of remotely located research stations of ICAR and SAUs.

Summer / winter schools: The Education Division of ICAR sponsors summer / winter schools with a view to orient the scientists and teachers on the recent advances in the field of agricultural research and education management.

International programmes: NAARM is being looked upon by the developing countries to provide training support for their human resource development in the areas of agricultural research and education management. The Academy has developed expertise and excellent facilities to cater to these increased demands from the NARS of various developing countries.

During the reported period, the Academy organized a total of **45** training programmes and trained **1028** personnel belonging to scientific, technical and administrative cadres of National Agricultural Research System in India and other developing countries.

NATP sponsored training programmes

1. **Retreat programme for top executives of ICAR:** Through an institutional consultancy arrangement, this programme was designed through IIM, Ahmedabad. The phase I of the programme was organized at

IIM, Ahmedabad during June 8 - 10, 2001. Eighteen senior level functionaries attended the programme. Finalization of the dates of Phase II of the programme is in progress.

2. **Faculty development at NAARM:** A detailed year-wise training schedule for the NAARM faculty was prepared and submitted to NATP for approval.

3. **Institutional development plan**: The institutional development plan prepared for seeking additional NATP support to NAARM was further modified by incorporating the suggestions made by the World Bank and NATP officials. The modified plan was further reviewed in a meeting of the high level committee appointed by NATP on January 10, 2002 at NAARM.

AHRD sponsored training programmes

Agricultural Human Resource Development Project, operating at NAARM, is focussed at developing 'educational technology cells', strengthening of existing facilities at the Academy and conducting teacher improvement programmes for various agricultural universities. Under this project, extensive training to sensitize the Professors, Associate Professors and Assistant Professors (numbering 162) of TNAU, Coimbatore on Computer Aided Instructional Technology was imparted. In addition, 109 administrative functionaries were also given training on administrative management and computer applications. During the period, Senior Executives Computer Lab and Executive Committee Room were developed. The final evaluation report of AHRD operated at the Academy was prepared.

International programmes

1. Programme on agricultural research management for scientists of Sri Lanka

On the basis of positive feedback received from the scientists and administrators from Sri Lanka, who participated in three training programmes organized at NAARM under the CARP-ICAR Work Plan, a special request was made to NAARM by the CARP to design and offer an international programme in agricultural research management in Sri Lanka for their agricultural scientists. With the approval of DARE/ICAR and the support of National Science Foundation (NSF) in Sri Lanka, a three-day programme was designed and offered during September 3 to 5, 2001 at the Postgraduate Institute of Agriculture, University of Peradeniya, Sri Lanka. Thirty-nine scientists from various research establishments in Sri Lanka attended the programme and got benefited. The programme as a whole was well received by the participants as well as the sponsors (CARP and NSF).

2. Programme on agricultural research management for scientist of Republic of Yemen

As a follow-up of the successful organization of a specialized two-month training programme in agricultural research management for two Research Directors from the Republic of Yemen at NAARM in 2000, the Academy was once again requested by the FAO to organize a similar programme for one senior level scientist for three months. With the concurrence of DARE/ICAR, a specialized training programme in English Language (Part 1) and Agricultural Research Management (Part 2) was developed, and is being offered from March 29, 2002.

National Programmes

Foundation Course for Agricultural Research Service (FOCARS)

73 FOCARS June 1 to Sept. 28, 2001 **74 FOCARS** Oct. 16, 2001 to Feb. 12, 2002	The FOCARS is the basic induction training imparted to all the newly recruited scientists to the agricultural research service. This four-month duration course is conducted in three specific phases namely, Phases I: Agricultural Research Systems and Management Processes - Concepts & Principles; Phase II: Field Experience Training and Phase III: Agricultural Research Systems and Management Processes - Practice and Applications. Phases I & III are organized at the Academy, and Phase II is conducted at the FET centres. The foundation course aim at exposing the participants to the various aspects of agricultural research project management, in addition to developing skills in them for managing their research by inculcating interdisciplinary team spirit. Adequate opportunities were provided to the scientists to develop knowledge and skills in information technology, communication, organizational behaviour and administration and finance. This training also aims at developing a feeling of fraternity among the newly recruited scientists. Various topics included Scenario analysis, research project management, organizational behaviour, individual and group behaviour, team building, computer applications & information management, technology transfer, educational technology including video production, administration and financial norms, and scientific communication. NAARM teams of Drs M Narayana Reddy, R Kalpana Sastry, VKJR Rao, D Rama Rao, B.S. Chandel, and K.H. Rao coordinated the foundation courses organized during the year. A total of 55 newly recruited scientists were trained through these two courses.

Refresher Courses

Information Technology in Agriculture July 10 to 30 and Dec. 3 to 23, 2001	Two programmes were organized to review and enhance existing knowledge of computer applications in agriculture and make effective use of Internet and information systems technologies in information exchange and management support. Fifty-five scientists and faculty of ICAR and SAUs with basic knowledge of computers attended the programmes. Drs N.H. Rao, M Narayana Reddy, K. Vidyasagar Rao, B.S. Sontakki and S.K. Soam coordinated these programmes.
Recent Advances in Agricultural Research Project Management Sept. 10 to 30, 2001	The programme was organized to equip the participants with knowledge and skills in various aspects of agricultural research project management. The programme dealt in detail the agricultural scenario, research project management, PRA techniques, information technology, scientific communication, human resource management, WTA & IPR. Nineteen scientists from ICAR institutes and SAUs attended the programme. Drs D. Rama Rao and M.M. Anwer coordinated the programme.

Senior Programmes

Designing and Developing Web pages for the Agricultural Technology Information Centres April 16 to 21, 2001	The programme was organized with an objective to study relevant web sites for information and use in web designing. The programme dealt with Creating web pages, applying themes, formatting text, creating internal and external links, creating navigation bar, inserting images, hyper linking images, creating tables, enhancing tables, adding web extras using the Front Page. Selected extension and education web sites and Agricultural Gateway of India (AGI)/ Agricultural Technology Information Centre (ATIC) prototypes were demonstrated. Twelve scientists from the ATIC centres participated in the course, coordinated by Drs N. Sandhya Shenoy and M. Narayana Reddy
Training of Trainers April 17 to 28, 2001	The concept and use of adult learning and experiential learning principles, coupled with the use of advanced training methods, are bound to increase the training effectiveness. This programme helped to orient the trainees to the concept and use of the above ingredients in training, with a view to improve their skills and effectiveness. The programme covered various important aspects of training, planning and management, through an experiential and co-training approach, that facilitated the trainees to see for themselves the effectiveness of these approaches, thereby motivating them to adapt these ideas and methods for use in their back home situation. The programme was coordinated by Drs T Balaguru and P Manikandan.
Management Development Programme in Agricultural Research (5th & 6th) April 19 to 25 and Nov. 21 to 27, 2001	It was emphasized in different forums that the Academy need to organize Management Development Programmes (MDP) in Agricultural Research for the newly-recruited Heads of Departments and Project Coordinators of ICAR Headquarters and various research institutes. Keeping in view the management responsibilities of these research managers, the fifth and sixth MDPs in Agricultural Research were organized at the Academy with an aim to provide a broad overview on the various dimensions of research management, human resource management, information management, and financial management. Heads of Departments and Project Coordinators numbering twenty participated in these programme, which were coordinated by Jagannadham Challa, S.N Saha and RVS Rao. 
Agricultural Scientist Development for Personal and Organizational Effectiveness June 12 to 23, 2001	The objectives of this programme were to examine different organizational structures, cultures and processes, and to get acquainted with organizational information systems and technologies and to adapt the model and approaches of the behavioural sciences in their back-home situations. The programme dealt with HRD strategies for agricultural research, personal effectiveness attributes, interpersonal relationships and transactional analysis, management of work stress, performance appraisal, management of creativity and problem solving,

	group behaviour and group dynamics project, management of differences, information systems, effective communication, vigilance and disciplinary procedures, and team management. Eight scientists / faculty from ICAR Institutes and from SAUs participated in the programme. Drs S. Shanmugam and Jagannadham Challa coordinated the programme.
Instructional Aids for Effective Presentations June 18 to 23, 2001	In the present era of communication and information explosion, the information management and its effective presentation holds the key for success. The instructional aids help in vitalizing presentations, attract and sustain interest, provide variety, improve understanding of complex concepts, promotes organized presentation, simultaneity of contact and help in reaching more people. Therefore, this programme was organized with the objective to impart knowledge and skills in planning, production and use of instructional aids for effective presentations. Drs N Sandhya Shenoy and R Kalpana Sastry coordinated the programme.
Agricultural Research Prioritization Techniques June 25 to 30, 2001	The programme was organized with the objectives to familiarize the scientists with the various techniques used for prioritizing agricultural research projects and to share with them the practical experiences of priority setting at different levels. It dealt with the policy context and need for research prioritization, participatory rural appraisal methods to understand the problems faced by the farmers, and the techniques of short listing and prioritizing the research problems. Twenty-seven scientists from ICAR institutes and SAUs, who are connected with the work of agricultural research prioritization and those connected with the preparation of strategic research extension plans, participated in the course. Drs K.P.C. Rao and B. S. Chandel coordinated this course.
Organizational and Management (O&M) Reforms in Administrative and Financial Management July 23 to 31, 2001	The programme was organized to improve the skills and efficiency of the Administrative and Finance Officers of ICAR in discharging their functions and providing constructive administrative support to Directors / Project Directors in meeting the organizational priorities and needs. It dealt with overview of ICAR society, reservation policy, pay fixation, retirement benefits, foreign deputation, purchase procedures in ICAR system, effective management strategies to improve administrative efficiency. Thirty-four Administrative Officers, Finance and Accounts Officers, Assistant Administrative Officers, Assistant Finance and Accounts Officers of ICAR institutes and Section Officers of ICAR Headquarters attended the programme. Mr. Suresh Kumar, Mr S.K. Pathak and Dr B.S. Sontakki coordinated the programme.
Analytical Tools and Techniques for Assessing Agricultural Sustainability Aug. 1 to 10, 2001	The programme was organized to introduce the various analytical tools and methods to define, quantify and assess sustainability of agricultural production systems and to provide hands-on training in the use of above tools and methods. The programme dealt with concepts of sustainable agriculture; sustainability at different hierarchical levels of agricultural systems; directions of required paradigm shifts in agricultural research, remote sensing, natural resources database management and GIS. Twelve scientists and faculty of ICAR and SAUs with basic knowledge of computers attended the programme. Drs M. Narayana Reddy and N. H Rao coordinated the programme.

Computer Applications in Agriculture Aug. 20 to 31,2001 and Feb. 5 to 15, 2002	The developments in computer science and information technology have profound impact on all the sectors of science and technology. Application of computers in agricultural research and development has, therefore, become more relevant and important to-day than ever before. Every scientist in the National Agricultural Research System needs to possess the knowledge and skill in working with computers. The Academy offers training programmems in computers and information technology to help achieve this goal. These programmes dealt with concepts of windows, word processing, spreadsheets, DBMS, presentation techniques, statistical analysis, networks, and multimedia. A total of twenty-six scientists, faculty members and technical officers from ICAR institutes, SAUs and other related organizations participated in these two programmes. Drs K.M. Reddy and K Vidyasagar Rao coordinated these programmes.
Impact Assessment of Agricultural R&D Sept. 17 to 22, 2001	The programme was organized to provide an overview and description of the major methodologies involved in impact assessment in different areas of agricultural research and development and to discuss empirical studies and experiences in application of impact assessment techniques. The programme dealt with techniques of R & D impact assessment, impact assessment of different agricultural technologies, assessing spill-ver and spill-in effects of agricultural technologies, issues in impact assessment of resource management and policy research. Ten research scientists of middle and senior level from ICAR institutes and SAUs participated in the programme. Drs B. S Chandel and K. P.C. Rao coordinated the programme.
Stress Management Oct. 17 to 24, 2001	Stress management is gaining more and more importance now a days, particularly in the organizational context. Scientific job is not a stress-free job. Scientists, on their work are exposed to tension, frustration and anxiety while executing their duties assigned. First time the Academy has organized this type of programme for the benefit of the scientists and faculty members of ICAR and SAUs. The programme gives an insight to the participants about the role of stress and coping mechanisms. It contains several instruments, exercises and case studies to measure the role-related stress of the individual scientists and also facilitates them to develop appropriate coping mechanisms and strategies. Four scientists attended the programme. Drs S. Shanmugam and P. Manikandan coordinated the programme.
GIS Applications in Agricultural Research Oct. 30 to Nov. 9, 2001	Geographical Information Systems (GIS) has proved to be a versatile tool in recent years in dealing with spatial data for realistic understanding and efficient management agricultural processes. The application of GIS in agricultural research, management and extension are growing. Several institutions of National Agricultural Research System have recognized the importance of GIS and have invested or propose to invest in establishing GIS facilities. There is growing need to build the capacity of NARS in making effective use of GIS based technologies in agricultural research and management. The Academy has organized this programme to introduce the relevant definitions and concepts of geographical information systems and related technologies like remote sensing. The programme dealt with map projections and scales, vector digitization of maps, spatial analysis, GIS and remote sensing, planning and implementation of GIS. Eight scientists and faculty of ICAR and SAUs with basic knowledge of computers attended the programme. Drs N.H. Rao and M. Narayana Reddy coordinated the programme.

Advanced Data Analytical Techniques for Agricultural Research and Management Nov. 19 to 29, 2001	This programme was organized to familiarize and create confidence among the trainees to analyze the data using various computer interactive analytical softwares such as statistical, graphical and simulation modeling and provide state-of-art analytical tools to 'optimize the information'. Two scientists / faculty of ICAR/SAUs with basic knowledge of computers participated in the programme, which was coordinated by Drs M.N. Reddy and K. Vidyasagar Rao.
Development Oriented Research in Agriculture Dec.10-15, 2001	Development oriented research in agriculture means the agricultural research, which takes systems approach, involves all stakeholders in decision making at various levels and is conducted by interdisciplinary team. Many times the desired change in farming situation is not achieved due to lack of understanding of relevant systems, non-involvement of key and other stakeholders, and insufficient analysis of natural and socio-economic circumstances of farmers. Therefore, the programme was organized to enable the participants to understand and analyze agricultural system and its properties and to equip them with the tools which help in translating research needs into plan of action, and prioritizing and field level options for innovation. Ten scientists and faculty of ICAR and SAUs attended the programme. Drs S. K Soam and B. S. Sontakki coordinated the programme.
Executive Development Programme in Agricultural Research Management Dec.21-24, 2001	The programme aimed at examining the challenges of general institute manageament, information management, and organizational effectiveness. The programme covered various themes, viz. institute management, administration management, financial management, stress management, auditing and accountability, vigilance and disciplinary procedures, leadership development and values and ethics in agricultural research. Fourteen Directors participated in the programme, coordinated by Drs Jagannadham Challa and S.N. Saha.
Refresher Course on Administrative and Financial Management Jan. 16 to 24, 2002	The programme enabled the participants to develop an appreciation and understanding of the various issues of administrative and financial management in the ICAR system, with a view to enable them to work positively and in team to achieve the organizational goals. Administrative officers and Finance and Accounts officers, numbering twenty-five attended the programme. Mr M. Suresh Kumar and Mr S.K. Pathak coordinated the programme.
Methodologies for Manpower Planning in Agriculture Jan. 17 to 19, 2002	The programme was organized to orient the resource persons of NATP project on manpower forecasting in agriculture and to develop appropriate tools and techniques for assessment and forecasting of trained manpower in agriculture. Fourteen scientists and faculty from NARS participated in the programme. Drs D. Rama Rao and S.K. Nanda coordinated the programme.
Project Management Using Microsoft Project Mar. 4 to 8, 2002	This course aimed at developing the knowledge and skills of participants in using Microsoft Project for efficient and effective research project management. It covered the concept and tools of research project management with emphasis on MS project tools in effective monitoring of the project resources. Six participants participated in this course, which was coordinated by Drs S.K. Nanda, N.H. Rao, and D. Rama Rao.

Sponsored Programmes

Sensitization Training Programme on Computer Aided Instructional Technology for TNAU Teachers April 10 to12, May 28 to 30 and Nov. 8 to 10, 2001	To sensitize the senior faculty members of the Tamil Nadu Agricultural University (TNAU), Coimbatore, to the basic concepts of computer and its applications, and to develop necessary skills in them in the use of computers for improving the quality of agricultural education, three sponsored, tailor-made programmes on Computer Aided Instructional Technology were specially organized for the senior-level faculty members of TNAU. The participants were provided with adequate hands-on practice with computers to develop skills in Word Processing, Electronic Spreadsheet, Database Management, and Slide preparation and presentation, in addition to exposure on Internet and e-mail. In all eighty-one senior faculty members were benefited by participation in these programmes. Drs A. Gopalam, R .V. S. Rao, and S. K. Soam coordinated the first programme. The second programme was coordinated by Drs.T. Balaguru, S.K. Nanda, and B.S. Sontakki. Drs P. Manikandan, M. N. Reddy and S.K. Soam were the coordinators of the third programme.
Computer Based Instructional Technology for TNAU Faculty April 26 to May 2, June 26 to July 2 and Nov. 19 to 25, 2001	Applications of computer in day-to-day activities of personnel engaged in agricultural research and development is more relevant today than ever before. Introduction of graphical user interface 'Windows' in the 90s made computers more user friendly than earlier. This programme is specially designed to expose the participants from Tamil Nadu Agricultural University to the facilities offered by MS Office and other packages and avenues of their application in their work. The programme dealt with MS Word, MS Powerpoint, Front-page overview and creating links, internet and interactive learning and research, and creating electronic books. In all eighty-one senior faculty members were benefited by participation in these programmes. Drs K. Vidyasagar Rao, M.M. Anwer, R. Kalpana Sastry coordinated the first programme. The second programme was coordinated by Drs N.H. Rao, K.M. Reddy and S.K. Soam. Drs K.P.C. Rao, K.H. Rao and A. Gopalam coordinated the third programme.
Modern Management of Administration for the Administrative and Accounts Officers of TNAU May 17 to 19, 2001	This programme was specially and exclusively designed for the functionaries of administration and finance wings of TNAU to achieve total quality. Various dimensions related to agricultural education, economic reforms, personnel and financial management, personality development and other general administrative and financial issued formed the curriculum of this programme. NAARM faculty Mr. M. Suresh Kumar, Mr. B. Pandu Reddy and Dr N. Sandhya Shenoy coordinated this programme in which twenty-six personnel belonging to the administrative and finance wings of TNAU were trained.
Computer Applications for Agricultural Research and Education for Administrative Staff of TNAU Aug. 2 to 8 Dec. 10 to 16 and Dec. 18 to 24, 2001	The developments in computer science and information technology have profound impact on all the sectors of science and technology. Application of computers in agricultural research and education has, therefore, become more relevant and important to-day than ever before. Every administrative and accounts staff needs to possess the knowledge and skill sin working with computers. To sensitize the administrative and accounts staff of the TNAU to the basic concepts of computers and its application, and to develop necessary skills in them in the use of computers for improving the performance, three tailor-made programmes were specially organized for those personnel. These programmes dealt with concepts of windows, word processing, spreadsheets, networks,

	Internet, e-mail, and multimedia. A total of eighty-three members participated in these three programmes. Dr M.N. Reddy coordinated the first programme. The second programme was coordinated by Drs R. Kalpana Sastry, N.H. Rao, and B.S. Chandel. Dr T. Balaguru, Mr. M. Suresh Kumar and Mr S.K. Pathak coordinated the third programme.
Programme on Educational Psycho-logy and Leadership Development for the Faculty of TNAU Dec. 27 to 29, 2001	Agricultural education perspectives need to be reoriented to meet the changing demands and challenges. There is, therefore, a definite need to bring about attitudinal and behavioural changes among the faculty members imparting agricultural education. They need to be updated with needed skills to meet the challenges in agricultural education. Keeping this in view, this programme was organized, specially for the faculty members of the Tamil Nadu Agricultural University, Coimbatore. Fifty-four faculty members attended this programme. Drs A. Goplam and S. N. Saha coordinated the programme.

Off-campus Programmes

Retreat Programme for ICAR Executives at IIM, Ahmedabad June 8-10, 2001	In view of experience and expertise available with IIM, Ahmedabad in the specialized area of institution building through leadership development, an 'Institutional Consultancy' was sought to design and organized a Retreat Programme for the Top 20 Executives of ICAR. The main objective was to sensitize the top 20 leaders to the macro economic transformations occurring in the global scenario with special emphasis on agriculture and to identify and understand the challenges confronted by ICAR in achieving its Vision 2020. Eighteen senior level officers of ICAR participated in the programme. NAARM faculty Drs J.C. Katyal, T. Balaguru, and Prof. Indira Parikh, IIM, Ahmedabad coordinated the programme.
Training Programme on Agricultural Research Management for Sri Lankan Scientists at the Post Graduate Institute of Agriculture, University of Peradeniya, SriLanka Sept. 3 to 5, 2001	The programme was organized for the scientists of Sri Lanka. The major objectives of the programme were to expose the participants to various principles of agricultural research management and to analyze and synthesize Indian experience on specific areas of management by the participants for immediate application in their country. Thirty-nine scientists from various research establishments in Sri Lanka attended the programme and got benefited. The programme was coordinated by Drs J.C. Katyal, and T. Balaguru.

Feedback

The feedback loop is an important mechanism, which makes the training programmes dynamic in nature. One of the striking features of the programmes organized by the Academy is feedback on various aspects of the programmes, gathered through a structured questionnaire. Comments / suggestions, thus received, are collated and synthesized into recommendations for revising prevalent methodologies/approaches on organization and delivery of various courses to introduce future improvements. The following table summarizes the feedback received on various programmes, the strengths being listed under 'evaluative' and suggestions for improvement being listed as 'corrective'.

Programme	Feedback	
	Evaluative	Corrective
FOCARS	- Overall content and organization of the programme very good and useful - Skills gained in specific areas for becoming a better researcher	- More time needed for computer practicals - More time needed for coverage of project proposal preparation - More guest lectures of outside resource persons needed - Need to reduce the number of theoretical lectures - Criteria for FET group formation need a relook
Senior programmes	- Good educational experience - Relevance and need of the programme - Acquisition of new knowledge and skills - Usefulness of exercise sessions and hands-on-practice sessions - Quality resource material was supplied	- Duration of these programmes should be increased - More time needed for computer practices - Refresher courses may be planned in the future to other colleagues - Tighter schedule of the programme - Case studies needed - Educational visits may be arranged
Sponsored programmes	- Useful and good study material provided - Informal and non-informal learning from the programmes	- More case studies and exercises need to be included to ensure variety - Presentation of some topics needs improvement

	- Commitment and expertise of the faculty members - Course content, modus operandi, and practical utility of the programmes very satisfactory	- Need for involvement of the trainees in every lecture - Resource material of the guest speakers are not provided - Need more time for practicals - Time duration of different programmes need a reconsideration - Need for more study visits to institutions in and around - More guest lectures could have been arranged
Meeting / Workshops	- Usefulness of group discussions and exercises - Effective use of case studies in the programme - Relevant to the needs - Provided a forum for sharing information on topical issues	- More case studies need to be used - Inadequate representation of top level management functionaries in the programmes

Research Highlights

The Academy scientists conduct research in the areas of agricultural policy, research project management, transfer of technology, human resources development, agricultural education, and information technology. Funding for some of the projects is through AP Cess, DFID and NATP. In addition, the Academy also supports short-term projects aimed at developing specific case studies for use in its programmes. A list of the projects carried out during 2001-02 is detailed below:

S No	Project Title	Funded by	Investigators
1	Management parameters of state agricultural universities and emerging policy issues and implications	AP Cess	A. Gopalam S. N. Saha
2	Developing instructional modules for gender in agriculture curriculum	AP Cess	A. Gopalam S. N. Saha
3	HRD strategies for organizational effectiveness in NARS	AP Cess	Jagannadham Challa K. Vidyasagar Rao
4	Research priority setting at micro level: Strategic Research and Extension Plan (SREP)	NATP	K. P. C. Rao, B. S. Chandel, N. Sandhya Shenoy, S. K. Soam, V. K. J.R.Rao, B. S. Sontakki
5	Performance assessment and accountability enhancement of Indian NAROs	NATP	T. Balaguru R. Kalpana Sastry R.V.S.Rao
6	Forecasting trained agricultural manpower	NATP	D. Rama Rao S. Shanmugam S.K. Nanda
7	Development of computer mediated courseware for facilitating learning – teaching	NATP	A. Gopalam K.M. Reddy
8	Capacity building of NARS in developing decision support system using GIS	NATP	N. H. Rao M.N. Reddy B.S. Chandel S.K. Soam and K.H. Rao
9	Development of virtual campus for agricultural research and education management	NATP	K. Vidyasagar Rao N.H. Rao M.N. Reddy K.M. Reddy

10	Distance training for agricultural research management (In collaboration with ISNAR, COL & Wye College)	DFID	Jagannadham Challa Dr. Byron Mook, ISNAR S.K. Soam B.S. Sontakki V.K.J.R. Rao
11	Impact assessment of oilseed research in India	NAARM	B.S. Chandel
12	Role of government and non government organization run Krishi Vigyan Kendras in agricultural development with respect to A.P.	NAARM	N. Sandhya Shenoy K. Vidyasagar Rao
13	Implementation of farming systems research in India:Case studies	NAARM	T. Balaguru P. Manikandan

Major activities and accomplishments of the projects during the year 2001-02 are briefly given below.

AP Cess fund projects

1. Management parameters of State Agricultural Universities and emerging policy issues and implications

Objectives:

- To study the ICAR model act and its implications in various agricultural universities.
- To probe into the acts, statutes and functioning of management bodies of SAUs.
- To ascertain the organizational structure of universities and their physical development.
- To investigate the integrated function of teaching, research and extension within the universities, and allied institutions within the state and the state departments of agriculture.
- To outline the course credit and internal assessment systems in various agricultural universities. To review investment pattern in SAUs and the outcomes.
- To scrutinize the recruitment and staffing pattern vis-à-vis the load of work and the outcome.
- To understand the planning, monitoring and evaluation mechanisms in the SAUs.
- To specify recommendations as a result of the above eight objectives and enlist the policy resulting out of this study.

Progress:

A comprehensive document both hard copy and CD was developed by developing a structured questionnaire, and was field validated for content criteria and purpose by piloting it to some selected respondents. An interview schedule was preceded for the research data in some selected universities.

2. Developing instructional modules for gender in agriculture curriculum

Background:

A research scheme with funding from A.P. Cess on developing instructional modules for integrating gender in agricultural curriculum was initiated during February 2001. The back ground of initiating such a research scheme is that urban graduates are reluctant to serve in rural areas, more number of rural students be admitted in the agricultural courses so as to increase the number of professional graduates with rural back ground, who may take up extension related jobs / activities. Women reservations are extended to all activities supported by state such as scholarships, admissions, and nominations to awards, curricular and co-curricular activities in order to ensure the gender equity in all developmental activities relating to agriculture. It is required that he organizational climate in the line departments needs to be made gender-friendly to encourage women to take up extension jobs and the women employees are to be provided necessary facilities and services like transport, accommodation and spouse employment, etc., in order to ensure more women participate in the over all social development.

It is proposed to use the instructional resource modules as courseware in Under Graduate instruction and thereby sensitize on the gender role in agriculture after initial validation and testing .The developed instructional modules will have the components for which adequate research work is contemplated.

Progress:

Three instructional modules were prepared on Gender roles, Gender resource management and Gender time management both in print and CD form. The developed modules were demonstrated to the teachers of TNAU and ANGRAU.

3. HRD strategies for organizational effectiveness in NARS

(This project was extended by one year upto May 31, 2002)

Objectives:

- To develop management systems approach for enhancement of organizational effectiveness.
- To develop and standardize case models for selected institutions for research productivity and personnel development of individuals.

Progress:

- The following databases have been prepared and analyzed: Management training needs assessment of scientists, Agricultural reasearch management practices and needs of ICAR Institutes, HRD climate in NARS, HRD needs assessment for administrative and finance functionaries of ICAR, Training database of NAARM programmes, Personality profiles of scientists and academicians.
- As a part of the project activities the following manuscripts are under preparation: HRD climate in NARS, Personality profiles of scientists and academicians and the final report of the project.

National Agricultural Technology Project (NATP)

Under the aegis of Organization and Management (O&M) Reforms component of National Agricultural Technology Project (NATP), two sub-projects, namely: (a) Institutionalization of Priority Setting, Monitoring and Evaluation and Networking of Social Scientists (to be implemented jointly by NAARM, NCAP and IASRI) (b) Human Resource Development for Agricultural Research and Education Management were sanctioned to NAARM for implementation during 2000-2003.

Major activities and accomplishments of the NATP projects during the year 2001-02 are briefly presented.

4. Research priority setting at micro level: Strategic Research and Extension Plan (SREP)

Objectives:

- To develop appropriate research priority setting methods at the micro level
- To train the scientists in micro-level priority setting

Progress:

- By utilizing the resource material developed by NAARM, two training programmes of one-week duration were organized during June 25 to 30, 2001 and September 17 to 22, 2001, which were attended by 37 scientists from ICAR Institutes and SAUs.

- In collaboration with MANAGE, the SREP methodology was refined and SREPs were prepared. The revised SREP guidelines are expected to be published soon by MANAGE.

5. Performance assessment and accountability enhancement of Indian National Agricultural Research Organizations (NAROs)

Objectives:

- To build NAARM's capacity in the area of organizational performance assessment and accountability enhancement
- To introduce innovative organizational performance assessment methods and processes for improving the performance of research institutions in ICAR

Progress:

- The team leader visited ISNAR and interacted with the members of the consultancy team associated with the project. A detailed report was prepared and a seminar was given to apprise the core team as well as other faculty members at NAARM on tools and techniques for performance of agricultural research organizations. Similar presentation was made in the PME Task Force meeting held in February 2002 at NCAP, New Delhi.

6. Forecasting trained agricultural manpower

Objectives:

- To design training programmes to forecast manpower needs in agriculture
- To provide policy guidelines for planning agricultural education so as to develop needed manpower in different occupations.

Progress:

- A three-day orientation programme was organized from January 17-19, 2002 at NAARM for the resource persons so as to share the project objectives, methodology and University-wise work plan.
- Some broad issues necessary for the project were identified. Sector-specific items were included in the questionnaires and they were mailed to resource persons for data collection.

7. Development of computer mediated courseware for effective learning - teaching

Objectives:

- To develop interactive learning modules in various agricultural subjects
- To develop educational technology learning modules for use in distance education

Progress:

- A seminar presentation was made and the opinions of the faculty were sought. The developed courseware was test administered to a group of learners in Tamil Nadu Agricultural University, through an Assistant Professor, to assess the efficacy of the software and the preliminary results are encouraging
- Using the Front Page Express and Hot Dog Professional and integrating U Lead 3 Cool Software, variety of instructional resource materials in Gender was developed.

8. Capacity building of NARS in developing decision support system using GIS

Objectives:

- To establish GIS Lab at NAARM
- To develop GIS based decision support system (DSS) for sustainable management of agricultural resources.
- To use the DSS as training tools for the scientists of NARS

Progress:

- Setting up of GIS Lab is in progress. Action for procurement of relevant equipment has been initiated.
- The state map of India and the Agroecological Sub-regions map have also been digitized in Geomedia GIS. The states map has been linked to All India State level database, and thematic maps of agricultural production, trends, etc. have been generated.
- A sample case study on research prioritization for rainfed ecosystems was developed using AGMIS and ARCVIEW GIS. A practical manual was developed for training.
- One training programme dealing exclusively with GIS applications in agriculture was organized from October 30 to November 9, 2001 covering the concepts, design, development and applications of GIS.

9. Development of virtual campus for agricultural research and education management

Objectives:

- To develop facility for establishing virtual learning centre
- To develop and registration of NAARM web site
- To make available relevant databases from the NAARM web site
- Design of home page and content module structures for virtual learning centre

Progress:

- The NAARM web site has been created and registered with the domain name icar.naarm.ernet.in and second domain name arm.naarm.ernet.in has also been registered for the virtual learning centre.
- Some areas like Research Project Management, Human Resource Development, Agricultural Technology Development, Educational Technology, Agricultural Related Databases were identified for the development of the centre.
- One of the topics of research project management namely 'Project management techniques -PERT' has been linked to NAARM web site on intranet and demonstrated the same to the trainee participants of Project Management. Using the Microsoft Project. The feedback received from the participants are being incorporated in the development of other lessons which are in progress.

ICAR / NAARM - ISNAR Collaborative Project

10. Distance training for agricultural research management

Objectives:

- The higher-level objective of this project is to enhance the contribution of National Agricultural Research System for sustainable rural development leading to the alleviation of rural poverty in India. Poverty alleviation is a national priority for India and a commitment of the CGIAR.
- The near-term objective is to improve the ability of the Indian NARS to focus agricultural research on poverty alleviation through distance training of researchers and agricultural research managers of Regional Research Stations of SAUs and ICAR institutes. This in-service distance training will focus on increasing conceptual awareness of poverty issues and on the development of relevant agricultural research management to target agricultural research needs of the rural poor. The project was implemented in two phases viz. applied research and adaptive research. A preferred strategy for distance training in agricultural research management was developed as an output of the applied researdh. This preferred strategy was adapted into a pilot distance training programme with reference to content, training strategy, learning strategy, packaging, delivery and support system.

During this period two more SAUs (TNAU and TANUVAS) were added to participate in the project and hence the pilot distance training programme was launched in five SAUs in November 2001.

Progress:

- The second edition of the communication directory of NARS was published and released in May 2001
- The communication directory was subsequently designed 'on line' and released through NAARM web page on July 13, 2001. The software for the Directory was prepared. This directory is now available on http//icar.naarm.ernet.in
- The distance training module on the theme "focusing agricultural research on poverty alleviation" was prepared and released for implementation in five SAUs - MPKV, Rahuri; ANGRAU, Hyderabad; UAS Dharwad; TNAU, Coimbatore and TANUVAS, Chennai.
- Training workhsop was conducted under the guidance of experts from Commonwealth of Lerning to orient and sensitize the site coordinators and nodal officers from five SAUs for launching of the pilot distance training programme (PDTP).
- Support system and feedback mechanisms were established to facilitate the PDTP.
- PDTP in 14 sites was launched in the first week of November 2001.
- Formative and summative evaluation of the programme is in progress.

Institute Projects / Case Studies

11. Impact assessment of oilseed research in India

Objectives:

- To estimate change in productivity and cost structure of oilseed crops
- To determine contribution of research and policy support to oilseeds productivity
- To evaluate return to investment of oilseed research in major oilseed crops in India.

Progress:

- The data collection on area, production and productivity was completed and analysis is in progress
- The results, which are expected, would reveal sustainability in total factory productivity and growth in TFP attributed to oilseed research in India.

12. Role of government and non-government organization run Krishi Vigyan Kendras in agricultural development with respect to A.P.

Objectives:

- To study, compare and assess the selected Organizational, Workgroup, and Job involvement variables and roles performed in agricultural development by the GO and NGO KVKs
- To note the expectation of the select clientele groups of the GO and NGO KVKs in view of changing agricultural scenario
- To find the inter-institutional linkage patterns among the NGO KVKs and other line departments, their expectations, operational constraints and suggestions for meaningful collaboration

Progress:

- The data collection from all the existing staff of the selected KVKs and the clientele groups was completed.
- The data have been compiled and the tabulation and analysis is under progress.

13. Implementation of farming systems research (FSR) in India: Case studies

Objectives:

- To investigate the FSR methodology followed by one research institution and to suggest suitable alternatives for effective implementation.
- To develop case studies that serve as resource material for training in FSR.

Progress:

- In order to understand the state-of-the-art of the implementation of farming systems research (FSR) approach in India, a case was developed based on the primary and secondary information collected on the Integrated Farming Systems Development Programme undertaken by the Agricultural Research Station of TNAU, Aruppukottai.
- The productivity, profitability and sustainability of the integrated farming systems developed by the scientists in the Research Station as well as those tested at the Farmers field were analyzed and described in the case study.
- The case study indicated an effective, efficient and a profitable integrated farming system (IFS) being followed by integrating the animal (mostly goats) and horticultural crops (fruit trees like Ber Custard apple) components in into cropping system. The yields of annual crops viz. sorghum increased by 20 to 25% under the IFS due to recycling of crop residues and got manure in rainfed areas. The IFS resulted in an additional income of Rs.7090 more than the normal cropping, over six years period. When the size of the holding is small, crop and animal (particularly goats) combination was found economical. In medium and large holdings, inclusion of goats/sheep, poultry and orchard crops was found economical. This has come due to increased and well-distributed income, as well as round the year employment offered by IDS, There is a need to develop multi-disciplinary working team for getting the maximum benefit from IFS.

Honours and Awards

Official language implementation award

The Academy won the Rajabhasha Puraskar of Town Official Language Implementation Committee, South Central Railway for the year 1999-2000 for the best work done in the implementation of the official language in twin cities. The award was presented by Shri N. Kirtivasan, General Manager, South Central Railway, at a special function held on June 26, 2001 at Railnilayam auditorium. Dr J.C. Katyal, Director, NAARM, received the award and briefed the heads of the departments of other organizations on the activities of NAARM with reference to official language implementation.

Lifetime achievement award

Dr J.C. Katyal, Director, NAARM was honoured with the lifetime achievement award for his long-standing research contributions in the area of micronutrients in agriculture. The award instituted by S. S. Ranade Memorial Trust, Pune, carries a cash prize, a silver plaque and a citation. Dr Katyal's basic research on the causes of and field research on the diagnosis for micronutrient deficiencies laid the ground on the necessity for micronutrient fertilization in sustainable production of food crops.

ICAR inter-zonal sports tournament

The NAARM sports contingent consisting of 36 sports men and women participated in the 18th ICAR (Zone-III) Inter-Institutional Tournament, held at CIFT, Cochin from December 4-9, 2001. The NAARM sports persons secured five individual and one team event prize. In the athletic events (women) Ms Rukmini Ammal bagged first prizes in shotput and discus throw. In the athletic events (Men) Mr M.K. Samson bagged two second prizes in shotput and javelin throw and third prize in discus throw. The volley ball team comprising of Mr M.K. Samson (Capt.), Dr Debnath, Mr C. Phani Raj, Mr Sham Bhadur, Mr P. Swamy, Mr C. Chandramouli, Mr N.R. Nageswara Rao, Mr Laxman Ahire, and Mr Shivaiah secured second place.

Again the NAARM Sports contingent, which participated in the final Inter-Zonal Tournament held at IARI, New Delhi from February 27 to March 3, 2002, secured five prizes. In the athletic events (Women) as in the previous tournaments, Ms Rukmini Ammal bagged the first prizes both in shot put and discus throw. The individual championship was also won by Ms Rukmini Ammal among women. In the athletic events (men) Mr M.K. Samson bagged the third prize in shot put and second prize in javelin throw.

NAARM feted at Rose Show

The Academy bagged the Hyderabad Rose Society Challenge Cup, Deccan Chronicle Trophy and T.Chandrasekhar Reddy Memorial Tropy at the 26th Annual Rose Show held at the Horticulture Centre at Public Gardens on December 9, 2001. The show was organized by the Hyderabad Rose Society. As many as 22 participants - individual and institutional - took part in the Rose Show, which was inaugurated by the Hon'ble Agriculture Minister of Govt. of Andhra Pradesh Vadde Sobhanadeeshwara Rao. The Academy also bagged the Sigma trophy for the best red rose of the show and Challenge Cup for the best display of flowers by Public Institution. Apart from the above the Academy won 13 first prizes, 15 second prizes and 9 third prizes under different sections of flower display. Again in the XII A.P.Horticultural Show organized by A.P. Horticultural Department at Public Gardens on January 26 to 28, 2002, the Academy bagged nine first prizes and ten second prizes.

Falicitation to Dr Panjab Singh

"There is a need to look at the new options to meet the existing and emerging challenges" said Dr Panjab Singh on the occasion of his visit to the Academy on October 10, 2001 after assuming charge as Secretary, Department of Agricultural Research and Education (DARE), and Director General, Indian Council of Agricultural Research (ICAR), New Delhi. Change effected after weighing all pros and cons is a path to growth. " The dynamic institutions have grown through needed periodic reforms in order to meet the new challenges over time. The Indian Agricultural Research System must also gear itself to address the new challenges. The human resources development is a necessary commitment of all dynamic systems for stability and attaining equilibrium with external forces ", he added. He intended NAARM to inculcate team spirit to ensure academic excellence and strive for overall personality development of agricultural scientists in the ICAR system.

Earlier, Dr J.C. Katyal, Director, NAARM, welcomed the gathering and felicitated Dr Panjab Singh and said that Dr Singh is an able research administrator with determination and total commitment. He has been successful in bringing about some positive changes in ICAR system he added.

Workshops, Conferences, Meetings and Summer Schools

Workshop on Motivational Techniques

The ability to motivate is an important skill in human resources management. Motivation is the non-monetary and non-tangible input in the organization, which tends to go a long way in improving the organizational efficiency and effectiveness and in providing inspiration and encouragement to the people to put in their best. Research Managers need to hone their skills of motivating the knowledge workers and becoming adept at employing the latest motivation techniques. The Workshop on Motivational Techniques was organized from April 10 to 12, 2001 to survey the classical and modern information on motivation and to learn from the experience of the participants to forge a new agenda for motivation of scientists in the 21st century. Four project coordinators attended the workshop. Dr M.M. Anwer and Dr Janardhan Jee coordinated the workshop. Some of the recommendations brought out during the workshop include:

- Launch excellence by role modeling it. Be self-motivated.
- Build sense of commitment transform compulsion for performance into concern for performance. Build self-confidence and confidence in others.
- Support your people. Help but don't do for others what they should do for themselves. Make your subordinate feel valued.
- Know and understand the person you wish to motivate (spend time with and on him/her). Individualize your supervision.
- Practice transparent and open communication.
- Invest in HRD - Specific-sensitizing sessions / training for group dynamics, change in attitude, behaviour, perception, goal setting etc.
- Responsibility and authority should be delegated together. Involve more participation in decision making. Provide necessary autonomy. Offer choices to your worker.
- Set goals with moderate risk. Fix performance standard for each individual employee. Create opportunities for achievement.
- Choose right people for right job.

Summer School on Recent Advances in Agricultural Research Project Management

This summer school was organized from April 18 to May 8, 2001 to develop in the researchers future perspective of Indian agriculture, in response to changing global agricultural scenario and to acquaint them with appropriate tools and techniques for efficient and effective management of research projects.

The programme was conducted with a theme for each of the three weeks. The first week focussed on the agricultural scenario - present and future, and also the trends in agriculture towards sustainability. The tools for technology forecasting and using these for research management was also enumerated. Against this background, the elements of project management cycle were taught and this culminated in the preparation of project using MS Project software. Two-day hands-on session was also held for this exercise and the participants presented their work to the faculty. The third week concentrated on the building of teams, consequent inter-personal relations and management of conflicts between teams. Sessions on writing of research proposals, new topics like labour management, WTA, IPR were also dealt. The participants were given exercises for preparing good scientific proposals and then the presentations were discussed to suggest improvement. Twenty-one scientists from ICAR Institutes and SAUs attended the programme. Dr J.C. Katyal, was the Director of this summer school. Drs T. Balaguru, R. Kalpana Sastry, B.S. Sontakki and V.K.J.R.Rao coordinated the programme.

Summer School on Recent Advances in Instructional Technology

Developing an awareness of instructional technology not only frees teacher from their historic, primary function of content management but also improve the quality of agricultural education. Hence, there is a need to master knowledge and skills required for utilizing effectively the recent advances in instructional technologies, which should be applied in creating instruction system that will enable educators to acquire new knowledge and skills required to carry out the necessary functions of a radical transformations and improved system of educating. A summer school on recent advances in instructional technology was organized from May 10 to 30, 2001, with focus on teaching effectiveness and teaching excellence.

The topics covered in the summer school included management of agricultural education, instructional technology an overview, brain storming as a technique for learning teaching, motivation and demotivation in class room instruction, instructional video an overview, web based learning theory and practice, presentation of poasters as means of communication, communication strategies, class room management, communication strategy for behavioral modification, participatory learning in agricultural education, strategies and options, management of financial resources etc.

Twenty-five Assistant and Associate Professors of State Agricultural Universities and Deemed to be Universities attended the programme. Dr A. Gopalam was the Summer School Director.

Workshop on Awareness of Intellectual Property Rights in Indian Agriculture

The agricultural scientists should prepare themselves to meet challenges of emerging global intellectual property rights regime and to protect the country's rich bio-diversity and traditional knowledge said R Saha,, Director, Technology Information Forecasting and Assessment Council (TIFAC), while addressing the delegates at a day-long workshop on Awareness of Intellectual Property Rights in Indian agriculture on May 31, 2001. The workshop was organized jointly with the Patent Facilitating Centre (PFC) of Technology Information Forecasting and Assessment Council (TIFAC), Department of Science and Technology, New Delhi.

Dr J.C. Katyal, Director, NAARM said that intellectual input could be applied in scientific research to get patent applications filed for new inventions and improvements. "We must move fast to build required institutional mechanism and the competent human resource in order to protect the country's rich genetic resources and traditional knowledge he added. Mr M.K. Rao and Mr Shanti Kumar, Patent attorneys, and Mr N.R. Seth, Deputy Controller of Patents and Designs, Calcutta, also spoke at the Workshop. Dr R. Kalpana Sastry, Senior Scientist, coordinated the workshop.

Retreat Programme for Senior Executives of ICAR

A three-day Retreat Programme for the top 20 senior executives of ICAR was organized by the Academy, under NATP, at Indian Institute of Management, Ahmedabad from June 8 to 10, 2001 to sensitize them on macro economic transformations occurring in the global scenario with special emphasis on agriculture and the challenges confronted by ICAR in achieving its vision 2020.

Drs Indira J. Parikh, M.R Dixit and Bakul Dholakia, faculty from Indian Institute of Management handled the sessions. The conduct of the sessions was in an interactive mode, which encouraged active discussion and input from each of the participant. The team critically reviewed organizational efforts in ICAR.

Workshop on Gahan Hindi Prashikshan

Two Workshops on Gahan Hindi Prashikshan was organized from July 2 to 6, 2001 and February 4 to 8, 2002 for the employees of ICAR headquarters and ICAR institutes, who are working in the sections other

than Hindi. The major objective of the workshop was to discuss about the experiential usage of Hindi. More stress was given on practical exercises in order to enable the participants to learn the use of Hindi in discharging their official duties in Hindi.

The workshop exposed the participants to the official language implementation; responsibilities and expectations of Central Government Employees, different forms of correspondence, the difficulties faced by non-Hindi speaking employees in the implementation of Official Language Policy. Some tips to increase use of Hindi in the Central Government Offices etc. Forty-one members attended the workshop. The first workshop coordinated by Dr A. Gopalam and Mr Pradeep Singh and the second workshop coordinated by Dr A. Gopalam and Ms J. Renuka.

Workshop on Modalities of Implementation of Pilot Distance Training Programme

A three-day workshop on Modalities of Implementation of Pilot Distance Training Programme was organized from October 24 to 26, 2001 at the Academy as a part of the academy's project in collaboration with Commonwealth of Learning (COL), Canada and International Service for National Agricultural Research (ISNAR), the Netherlands. The workshop was organized for the benefit of the nominated nodal officers and site coordinators of the five participating universities (ANGRAU, Hyderabad; MPKV, Rahuri; UAS, Dharwad; TNAU, Coimbatore; and TANUVAS, Chennai) and to develop facilitation skills among them for providing support to learning activities, monitoring and evaluation of the pilot distance training programme.

The proceedings focussed on the role of site coordinators, learning issues, team building, logistics, module content, monitoring and evaluation of distance training manual. Dr Byron Mook, ISNAR, Dr Krishna Alluri, and Dr Helen Lentell, COL, 14 site coordinators from learning sites and five nodal officers from the five sate agricultural universities participated along with NAARM project team. Drs Jagannadham Challa, S.K. Soam, B.S. Sontakki, and V.K. Jayaraghavendra Rao coordinated the workshop.

Workshop on Priorities and Obligations of Official Language Policy

A four day workshop on Priorities and Obligations of Official Language Policy was organized from November 6 to 9, 2001 for the Hindi Officers, Asst. Directors (OL), Technical Officers (OL) Hindi Translators, Hindi Assistants, who are responsible for implementation of official language policy and rules in ICAR institutes.

The workshop focuses on implementation of official language policies and rules. A blend of lectures on the use and practical difficulties in the implementation of Official Language Policy of Government of India and possible solutions for them and lively discussions among the participants on various difficulties and future plans were among the high lights of the workshop. Based on their experience and as consequence of discussion, the participants gave various practical suggestions for better use of official language in their organizational functions. The topics covered during the workshop were priorities and obligations in implementation of official language policy in agricultural research institutes, various steps to be taken for effective implementation of official language policy, managerial and organizational difficulties in implementation of official language policy and rules, the contribution of parliamentary committees for efficient implementation of official language policy, motivating factors for appropriate compliance of official language policy in the interest of nation and presentations by the participants on various difficulties faced by them and their practical experience in overcoming them. Forty-seven members attended the workshop. Dr A. Gopalam and Mr D. Venkateswarlu coordinated the workshop.

Management Committee Meeting

The 35th Management Committee of NAARM met on November 24, 2001. The meeting was held under the chairmanship of Dr J.C. Katyal, Diretor NAARM and with the following members: Dr I.V. Subba Rao, Vice Chancellor, ANGRAU, Dr B. Mishra, Director, Directorate of Rice Research, Mr M. Suresh Kumar, CAO, NAARM and Mr S.K. Pathak, F&AO, NAARM. In a day-long session the committee discussed the progress report of NAARM for the period from September 2000 to October 2001 and on future programmes and activities.

QRT Meeting

The Quinquennial Review Team meetings were held from December 18 to 22, 2001 to review the Academy's organizational structure, physical facilities, financial matters, human resources in relation to overall national plans, policies and priorities; and to examine the policies, priorities, strategies and programmes undertaken by the Academy. The QRT team comprised of Drs H.K. Jain (Chairman), K.V.Raman, K. Pradhan, S.L. Mehta, and H. S. Nainawatee. Earlier, Dr K. Pradhan, visited the Academy from October 10 to 14, 2001 and interacted with the Director and the faculty members on the matters related to linkages and other collaborative institutes of the Academy. Dr K.V. Raman, visited the Academy from October 17 to 23, 2001 and interacted with the Director and faculty members on ongoing and proposed research activities. As the Member-Secretary, Dr T. Balaguru coordinated all these meetings.

Research Advisory Committee Meeting

The first meeting of the newly constituted Research Advisory Committee of the Academy was held on April 27, 2001. Dr J.C. Katyal, Director, NAARM accorded a warm welcome to the Chairman and Members of the newly constituted RAC of NAARM. For the benefit of the Committee, he made a brief presentation about NAARM covering various aspects of its organization and functions. He also highlighted the policy support provided from time to time by NAARM to ICAR towards realizing the improved efficiency and effectiveness.

In his opening remarks Dr H.K. Jain, Chairman, Research Advisory Committee, emphasized the important role that a unique institution like NAARM plays in brining about the required change and influencing the policy environment of a vast National Agricultural Research System (NARS) like ours. He highlighted the major challenges that the system is faced with on account of signing of the WTA. This opens up a whole range of management problems like attaining food security, making Indian agriculture globally competitive through quality improvement, increasing the profitability through reduction in cost of production, and ensuring the production and productivity gains to be economically viable, ecologically sustainable and socially supportable. This calls for greater responsibility on the part of NAARM to come out with appropriate strategies and policies for improving the management effectiveness of NARS.

While appreciating the mandate of NAARM as a very comprehensive one, he urged the Academy to identify priority areas of activity with the given set of resources and limit its programmes to only those areas that are of high priority. Dr S.K. Dutta, Chairman-CMA, IIM, Ahmedabad and Dr H.S. Nainawatee, ADG (HRD

II), ICAR, New Delhi attended the meeting. Dr T. Balaguru, Principal Scientist and Member-Secretary, RA(presented the report on the status of research activities and training programmes at the Academy.

Review of Performance Appraisal and Preparation of New Rules and Guidelines for ICAR

A one-day meeting on Review of Performance Appraisal and Preparation of New Rules and Guideline for ICAR was organized on January 25, 2002. Thirty-two scientists, administrative and finance officers fror local ICAR institutes and ICAR headquarters attended the meeting. Dr D. Rama Rao coordinated the meetinc

Dr T. Balaguru, Acting Director, NAARM, welcomed the delegates and initiated the discussions b highlighting contributions of NAARM in this area. Dr J.C.Katyal, DDG (Edn.) made brief presentation on th background to the preparation of the draft document and highlighted its broad features.

Mrs Sashi Mishra, Secretary, ICAR, New Delhi requested all the participants for comments on th document. The entire gathering reacted on every point made. Brief summary of the discussion is given below

- The Annual Assessment report need not be confidential and it should be an open one instead c confidential. This document should aim to build accountability and to minimize subjective assess ment by superiors.
- Though teaching and training are limited to few instiutes as most institutes carry research, it wa: agreed to have common format.
- Need for elaborate guidelines was felt or else it may lead to different interpretations by differen individuals
- A scientist may work in four projects and four team leaders may write the assessment. Assessmen may not be same for all projects. Then how to collate this.
- The form focuses on targets fixed and completed. Quality of work is not given much importance as i will be judged by peer group separately.
- Invitation to professional and special assignments by outside organizations and the parent organiza tion may be included
- If this is to be more effective, we require support system of Project Based Budgeting and delegation o powers to Principal Investigators.
- Team leaders for teaching, training and other activities may be identified at the beginning of the year
- Contribution to Institution/Organization to be included
- To eliminate bias and enhance objectivity in assessment the following weightage may be given: Scientific contribution 50%; Professional activities/behaviour 30%; Central behaviour 10% and any other (specify) 10%.
- For the whole document to be effective the Research Project File (RPF) needs to be matched o modified and integrated with the Annual Assessment Report
- Since subjectivity is involved while filling the remark column, this may be deleted.
- Space for training needs and potential appraisal is required in the document.
- Since the proposed assessment system is expected to be transparent and open, the scientist shoulc also be given an opportunity to make self-assessment on the attributes that are listed.
- Group suggested deleting of evaluation by team leader, as it does not improve the quality of assess ment in any way. Since the targets and achievements are clearly defined by the reviewee, which are also available in RPF documents and presented in SRC, the intervention of team leader is not essen tial. In the event of a scientist being associated in more than one project, the appraisal report has tc be sent to all the team leaders, which could bias the very evaluation of the scientist.
- No provision made for evaluation by the Reviewing Officer.

National Workshop on Change Management

The ability to manage change is an important skill in management of organizations. It is essential to develop this skill in order to succeed as a facilitator of knowledge workers. This is because the scientists are constantly working to expand the frontiers of science and create new knowledge. The application of this knowledge is bound to change the society in ways that was never envisaged before. Such creative organizations themselves have to be adept at changing themselves and the way they work. Management research has clearly established t hat those organizations and managers, who have learnt to effectively manage change, have progressed faster than others. There is a need to understand the forces that stimulate change and also to learn about the sources of individual and organizational resistance to change so that managing change can be successfully planned and practiced. Such a change will lead to organizational development and team building. A National Workshop on Change Management was organized at the Academy from March 14 to 16, 2002. This workshop aimed to survey the classical and modern information on management of change and also seeks to learn from the experience of the participants to forge a new agenda for change in the 21st century. There were four resource-input sessions, a structured experience for learning in change management and a session to generate the output of the workshop. The resource input session were 'Basics of change management', 'Approaches to managing organizational change, 'Strategies for change management'. Some of the action plans brought out during the workshop include:

- Changing Managers from MMM (Money Making Manager) to MOM (Mandate Oriented Manager)
- Relinquishing the scientists from non-scientific activities
- Changing the attitude of young scientists towards field-bound researchers (to tackle with computer - mania and desk-work tendency)
- Management of soil-borne diseases in pulses
- To improve the quality of agricultural research
- Agriculture in the context of WTO

Eight middle level managers of ICAR and SAUs who are project coordinators and Heads of Regional Research Stations participated in the workshop. Drs S.N. Saha, M.M. Anwer, and B. S. Sontakki coordinated this workshop.

National Seminar on Veterinary Drugs and Pharmaceuticals

A two-day National Seminar on Veterinary Drugs and Pharmaceuticals was organized on March 23 and 24, 2002 at the Academy. The Seminar was aimed to assess the status of Research and Development in the area of veterinary drugs and pharmaceuticals and to promote R & D collaboration between research institutions and industries dealing with veterinary drugs and pharmaceuticals.

'In order to meet out the challenge of providing nutritional / food security to the escalating human population, there is ever increasing demand on food producing animals and their products which can be achieved only through development of effective, eco-friendly, economical and safe drugs and pharmaceuticals that pave the way to alleviate the diseases and enhance the productivity and performance of animals" said Padma Vibhushan Prof. M.M. Sharma, FRS, while inaugurating the Seminar on Veterinary Drugs and Pharmaceuticals sponsored by Department of Science and Technology (DST), Government of India, at the Academy on March 23, 2002. Professor Sharma also said that though the animal population is more or less same as to human population in this country, the annual turn over of the animal pharmaceutical industry is very small (Rs.800 crores) compared to the human pharmaceutical industry (Rs.23, 000 crores). Indigenous scientific innovations have not yet made a significant impact in the area of veterinary drugs and pharmaceuticals. Much scientific effort was not made in identifying the compounds or molecules, which have therapeutic properties, and for refining and isolating them. As a result, our dependence on foreign companies for the supply of drugs to treat the animal ailments has continued to increase and also veterinary drug firms are using compounds used in human medicine to animal medicine, with the exception of rumen diseases. Why there is such disparity? What is the reason for this poor growth rate of animal pharmaceutical industry in our country? These questions are unanswered due to differences in our opinion. He also, stressed the need to know which of the drugs are being used in food producing animals, and also to know why, when and by whom these are being used. This information has become particularly more pertinent in the changing scenario of new industrial revolution and trade policy, he added.

Speaking on Department of Science and Technology's role in promoting Veterinary Drug Research, Dr Laxman Prasad, Advisor, DST, Government of India, mentioned that DST is already operating a Plan Scheme - Drugs and Pharmaceuticals Research Programme intervened to create and support an interest in R & D activity in the area of veterinary drugs and pharmaceuticals too.

Prof. Allauddin Ahmad, Ex-Deputy Director General (Edn.) and Former Vice Chancellor, Hamdard University, said that the research and development activities in the field of veterinary drugs are limited in India. Shortage of funds with State Agricultural Universities, market oriented approach of commercial organization, flaws in the registration procedures of veterinary drugs, are some of the factors which allowed stunted growth of veterinary pharmacy research and development.

Earlier Dr T. Balaguru, Acting Director, NAARM, welcomed the delegates, and thanked the Department of Science and Technology for honouring the Academy by choosing as a platform to organize this seminar.

The Seminar was organized in collaboration with Acharya N.G. Ranga Agricultural University (ANGRAU) and NAARM, Hyderabad. Sixty-four delegates working in the field of Veterinary Pharmacology, Toxicology and Veterinary Medicine in the State Agricultural / Veterinary Universities, ICAR Research Institutes and Veterinary Drug Firms from the Private Sector participated in the Seminar.

Dr D. Rama Rao, Head, Information and Communication Management Division, NAARM, and Dr K.S. Reddy, Professor and Head, Department of Pharmacology/Toxicology, Veterinary College, ANGURAU, Hyderabad coordinated the seminar.

Some of the recommendations brought out during the workshop include:

- A task force be formed by having representation from research institutions, industries and DST for follow up action.
- A National Institute of Veterinary Pharmacy, Pharmacology and Toxicology be established with regional centres. The main activity of this institute will be to coordinate the research development activities pertaining to veterinary pharmaceuticals.
- Information on expertise and facilities available is SAUs, Veterinary institutes and ICAR institutes and Industries may be gathered

- A separate directorate be established at the National level for approval, registration and regulatory function of veterinary drugs and pharmaceuticals including Aurvedic and Homeopathic veterinary medicine.
- A comparative review and critical assessment on the research work conducted in pharmacology and allied discipline may be made and recorded.
- Creation of facilities and infrastructure for synthesis of drugs which are being imported presently.
- Development of rapid diagnostic kits for detection of poisoning in animals.
- Detail studies on the development of fixed dose drug combination be taken up.

NAARM's Silver Jubilee Inaugural Function

The Academy's year-long Silver Jubilee Celebrations started with the lighting of a lamp by Dr M. Mahadevappa, Chairman, Agricultural Scientists Recruitment Board (ASRB), New Delhi, on September 1, 2001. Inaugurating the celebrations, Dr Mahadevappa said that the Academy occupies a unique place for its vision, activities and its continuing contribution to the National Agricultural Research System. Praising the services rendered by the Academy, Dr Mahadevappa expressed that the it could go into problems faced by scientists in the ICAR system. As a think tank for ICAR, it can evolve new policies for efficient governance and good management packages, which would bring efficiency and better output from scientists. He stressed on role of NAARM to inculcate team spirit and prepare scientists to face challenges posed by an increasingly technology driven world.

Dr J.C. Katyal, Director, NAARM, gave a bird's eye view of progress of the Academy over the last two and a half decades. Since inception, the Academy organized a series of training programmes, symposia, conferences and workshops, both at national and international levels. In order to further academic pursuits and simultaneously generate resources, the academy regularly undertakes consultancy work at the request of national and international agencies. The academy, he said, was intended to fulfil the felt-need of training scientists as better researchers.

Faculty news

Visits abroad

Dr T. Balaguru, Principal Scientist visited The Netherlands from March 31 to April 17, 2001 under the aegis of World Bank supported NATP Sub-project on Priority Setting, Monitoring and Evaluation (PME). Developed expertise in performance assessment and accountability enhancement of agricultural research organizations by interacting with senior officers at ISNAR, The Hague.

Drs J.C. Katyal, Director and T. Balaguru, Principal Scientist visited Sri Lanka from September 2 to 6, 2001 as a follow-up of the MoU signed between ICAR and Sri Lankan Council for Agricultural Research and Policy (CARP), organized an International Training Programme on Agricultural Research Management for the Sri Lankan scientists.

Dr Janardan Jee, Principal Scientist visited USA from July 1 to September 30, 2001 under the aegis of World Bank supported NATP Sub-project on Human Resource Development for Agricultural Research and Education Management. Developed expertise in human resource development and organizational development by interacting with senior faculty members at the Ohio State University, Columbus.

Participation in conferences, meetings, workshops, seminars, symposia, etc.

Dr P. Manikandan, Principal Scientist, participated in a programme on "IT Applications in Libraries", conducted by National Institute of Rural Development, Hyderabad from April 9 to 12, 2001.

Dr S.K. Soam, Senior Scientist, participated in the workshop on "National Assessment on Appropriate Technology Transfer for Rural Women" and functioned as a member of 'Committee on Disaster Management' at National Institute of Rural Development, Hyderabad from May 2 to 4, 2001.

Dr J.C. Katyal, Director and Dr M. Narayana Reddy, Principal Scientist, NAARM, attended the International Workshop on the Internet and Intranets, new opportunities for change in Agricultural Research and Education sponsored by ISNAR and JIRCAS at Chennai from June 11 to 16, 2001.

Dr B.S. Chandel, Senior Scientist, attended training program on 'Use of GeoMedia Professional Software' on June 15 and 16, 2001 at NAARM, Hyderabad.

Dr Jagannadham Challa, Principal Scientist, and Dr Byron Mook, Senior Research Officer, ISNAR visited TNAU, Coimbatore from June 15 to 17, 2001, and held discussions with participation of TNAU in the Pilot Distance Training Programme of NAARM and ISNAR training programme.

Dr N.H.Rao, Principal Scientist presented a paper on sustainable agriculture, Proc. of Workshop on Sustainable Agriculture, on July 13 and 14, 2001, Centre for World Solidarity, Hyderabad.

Dr N. Sandhya Shenoy, Senior Scientist, attended a National Seminar on " E-Learning and E-Learning Technologies" organized by C-DAC, Hyderabad on August 7and 8, 2001.

Dr N.H.Rao, Principal Scientist, presented a paper on GIS based decision support systems for groundwater management, in the First NATP Workshop on "Technologies for Skimming and Recharging Fresh Water in Saline and Groundwater Regions at Guntur", on Augst 9 and 10, 2001

Dr K.H. Rao, Senior Scientist, attended National Conference on "University-Industry-R&D Organisations-Scientific Bodies Interaction and Co-operation: Policy Perspectives" organised by Centre for Policy Reseach (CPR), National Institute of Research and Social Action (NIRSA) and Osmania University College of Engineering, Hyderabad, from August 11-12, 2001.

Dr S.K. Nanda, Senior Scientist, attended a workshop on Participatory Extension for Sustainable Natural Resource Management at National Institute of Rural Development, Hyderabad, from August 16 to 18, 2001.

Dr R. Kalpana Sastry, Senior Scientist and Dr N. Sandhya Shenoy, Senior Scientist, attended a National Seminar on "Gender Issues - Women in Agriculture and Management" organized by Tamil Nadu Agricultural University, Coimbatore from August 20 to 22 2001.

Dr R. Kalpana Sastry, Senior Scientist, attended a national workshop on "Challenges of liberalization and protection of farmers rights" at National Institute of Rural Development, Hyderabad on August 24 and 25, 2001.

Dr J.C. Katyal, Director and Dr T. Balaguru, Principal Scientist attended a National Workshop on "Agricultural Research Management" organized by National Science Foundation, Sri Lank from September 3 to 5, 2001.

Dr J.C. Katyal, Director, participated in the wrap-up meeting of the AHRDP on October 15, 2001 at ICAR, New Delhi.

Sr K. Vidyasagar Rao attended All India Agricultural Statisticians Conference held at Ludhiana from November 6 to 8, 2001 and presented a paper entitled "Computer Applications in Agriculture - Present and Future".

Dr S.K. Soam, Senior Scientist, participated in the national conference on "Agribusiness and Extension Management: Status, Issues and Strategies" at MANAGE, Hyderabad on December 7 and 8, 2001 and presented a paper entitled "Managing Agricultural Research: National Goal and Role of Agribusiness Organizations".

Dr J.C. Katyal, Director, attended the project review meeting to discuss World Bank AHRD - Implementation Completion Report and Final Mission 2001 on December 10, 2001 at ICAR, New Delhi.

Dr S.K. Soam, Senior Scientist, participated in the meeting on training component of "Network of Agri-clinics and Agribusiness centre scheme" at MANAGE, Hyderabad on December 22, 2001.

Dr S.K. Nanda, Senior Scientist, attended a Seminar on Rural Technology for Poverty Alleviation at National Institute of Rural Development, Hyderabad, on January 2 and 3, 2002.

Dr R. Kalpana Sastry, Senior Scientist, attended a International Seminar on "New Dimensions of Intellectual Property in Changing Scenario" at Hyderabad on January 24 and 25, 2002.

Dr B.S. Chandel and Dr S.K. Nanda, Senior Scientists attended the Technical Programme on Agricultural Projects - Planning and Appraisal at MANAGE, Hyderabad from February 10 to 12, 2002.

Dr R.V. Satyanarayana Rao attended a six-day training programme entitled "Effective communication aids for natural resource management" conducted by National Institute of Rural Development (NIRD), Rajendranagar from February 18 to 23, 2002.

Dr A. Debnath, Medical Officer attended the Institutional Biosafety Committee meeting at Directorate of Rice Research on February 27, 2002.

Dr S.K. Nanda, Senior Scientist, attended a workshop on Microsoft Project 2000 and Project Central at Centre for Excellence in Project Management (CEPM), New Delhi on March 5 and 6, 2002.

Dr M.N. Reddy, Principal Scientist, attended a training in using LIBSYS software procured by NAARM aat LIBSYS Corporation, Bangalore from March 11 to 20, 2002.

Dr Jagannadham Challa, Principal Scientist, attended the meeting conducted by the Secretary, ICAR on March 21, 2002 on Revision and Refinement of ICAR Rules held at Krishi Bhavan, New Delhi.

Invited lectures delivered

Dr N.H.Rao, Principal Scientist, was invited to be a member of ICAR-DARE working group for the Tenth FiveYear Plan on Agricultural Education held at Delhi, on May 8, 2001.

Dr M.M. Anwer, Principal Scientist, was invited as a gust speaker to deliver lectures to the participants of the training programmes of Water and Land Management Training and Research Institute of Govt. of Andhra Pradesh, Hyderabad on May 17, 2001, January 3, and 11, 2002.

Dr M.M. Anwer, Principal Scientist, was invited as a gust speaker to deliver lectures in interstate programmes conducted by the Extension Education Institute of the ANGRAU, Hyderabad on Junde 6, August 3, September 5, 2001 and February 6, 2002.

Dr R. Kalpana Sastry, Senior Scientist, delivered a lecture on "Training needs for farm resource management" to the participants of Training needs assessment for natural resource management at APARD, Hyderabad on June 13, 2001.

Dr S.K. Soam, Senior Scientist, delivered a lecture on "PRA techniques" at Punjab Agricultural Management and Extension Training Institute, Punjab Agricultural University, Ludhiana on August 10, 2001.

Dr M.N. Reddy, Principal Scientist, delivered a lecture on "Design of Experiments" to the participants of the Postgraduate Diploma Course in Plant Protection at National Plant Protection Training Centre, Hyderabad on September 7, 2001.

Dr J.C. Katyal, Director, attended the Indian Society of Soil Science 66th Annual Convention on October 29, 2001 and delivered 28th Dr R.V. Tamhane Memorial Lecture (Topic: Fertilizer use situation in India) on October 31, 2001 at Rajasthan College of Agriculture, Udaipur.

Dr T. Balaguru, Principal Scientist, was invited as a guest faculty at Extension Education Institute, (EEI), Acharya N.G. Ranga Agricultural University (ANGRAU), Hyderabad and delivered two lectures entitled " Concept and Importance of Management Information System (MIS)"and " Development and Management of Agricultural Information System (MAIS)" for the developmental personnel from the Southern States on November 21 and 22, 2001 respectively.

Dr S.K. Soam, Senior Scientist, delivered a lecture on "Project logical framework" at National Institute of Agricultural Extension Management, Hyderabad on November 23, 2001.

Dr N.H.Rao, Principal Scientist, was invited to brainstorming session for planning and implementation of a Citizen's Report on Agriculture organized by TIFAC and the Centre for Environment and Development in Delhi on December 12, 2001.

Dr R. Kalpana Sastry, Senior Scientist, was invited as a guest speaker in a panel discussion on "Crop Protection and WTO-an Indian Perspective" on January 23, 2002 and also presented on "IPR perspectives in Plant Pathology.

Dr R. Kalpana Sastry, Senior Scientist, participated in Diploma Programme on Organizational and Professional Ethics on Feb 21, 2002 at University of Hyderabad and took session on "IPR and Ethics".

Dr M.M. Anwer, Principal Scientist, was invited as the Judge in horticultural show organized by the Govt. of Andhra Pradesh, Hyderabad, on January 26, 2002.

Dr N.H.Rao, Principal Scientist, was invited to present keynote paper at First NATP workshop on Technologies for Skimming and Recharging Fresh Water in saline Aquifer Regions

Dr K. Vidyasagar Rao, Principal Scientist, was invited as a Guest Speaker by MANAGE to conduct a practical session on Statistical Data Analysis using MSTATC and SPSS to the participants of the training on Statistical Software's for Data Analysis on February 13, 2002.

Dr K. Vidyasagar Rao, Principal Scientist, was invited as a Resource Person by the Director, UGC – Academic Staff College, University of Hyderabad and delivered a lecture on "Design and sampling for social research" to the participants of need based refresher course in research methodology on March 18, 2002.

Dr K. Vidyasagar Rao, Principal Scientist, delivered a lecture on "Design of experiments" for the trainees of P.G. Diploma in Plant Protection, NPPTI, Hyderabad.

Dr K. Vidyasagar Rao, Principal Scientist, was invited as a member, Advisory committee for the Ph.D. students Mr Sunanda, G. Rajni Kumar and Mr Anji Prasad of agricultural economics of ANGRAU, Hyderabad.

Dr K. Vidyasagar Rao, Principal Scientist, took part in the comprehensive vivo-voce of the Ph.D. students of agricultural economics in ANGRAU, Hyderabad.

Dr B.S.Chandel, Senior Scientist, delivered lecture on "Micro-level Priority Setting" in Sensitization cum Training Workshop of Scientists of PME Cells in ICAR and SAUs' at National Centre for Agricultural Economics and Policy Research, New Delhi on January 2002.

Publications

Research / Review / Conference Papers

Bal Krishan Trivedi and **Sudhir Soam** 2002. Chara utpadan hetu beehad bhoomi ka prabandhan (in hindi). Rashtriya sangoshthi: warsha adharit krishi kshetro me vikratikaran niyantran, CRIDA, Hyderabad, March 5 to 7, 2002, 70p.

Bhattacharya, B and **Nanda, S.K.,** 2002. Socio-technical attributes of Shifting Cultivation in North-east India, Tui, Vol. X No.35, (March 2002), pp 7-22.

Chandel, B.S. 2001. Impact Assessment of Agricultural R & D Programme and Projects. Reading hand-out in Pilot Distance Training Programme on Agricultural Research Management.

Chandel, B.S. 2001. Socio-Economic Development of Watershed- A Project Management Approach. Indian Journal of Dryland Agriculture.

Chattopadhyay, C. and **Kalpana Sastry, R.** 2001. Integration of viable disease control tools for management of sesame stem-root rot caused by Macrophomina phaseolina. J. Mycol. Plant Pathol: 31 (in press).

Chattopadhyay, C. and **Kalpana Sastry, R.** 2001. Potential of soil solarization in reducing stem-root rot incidence and increasing seed yield of sesame. J. Mycol. Plant Pathol: 31 (2):227-231.

Kalpana Sastry, R. 2001. Towards labour policy in agricultural research institutions- special presentation to the administrative officers of ICAR headquarters. NPBGR Conference Hall, July 4, 2001 (concept paper submitted to ICAR vide D.O. letter Policy/2002 dated January 10, 2002).

Kalpana Sastry, R. and Chattopadhyay, C. 2001. Effect of soil solarization on wilt incidence and seed yield of safflower. J. Mycol. Plant Pathol.: (communicated).

Kalpana Sastry, R., Chattopadhyay, C., Vrijendra Singh and Hegde, D.M. 2001. Combining host resistance, cultural and chemical measures to manage safflower wilt. Indian Phytopathology (Presented at the IPS Central Zone Annual Meeting on 19 Dec 1998 at Dept. Botany, Osmania Univ., Hyderabad). J. Mycol. Pl. Pathol. (communicated).

Kalpana Sastry, R., Rama Rao, D., Kiresur, V., and Vijayakumari, R. 2001. Forecasting of safflower production in ndia. Seame and Safflower Newsletter, no.16:88-97.

Kalpana Sastry, R., and **Manikandan, P.** 2001. Case study of farm women empowerment in an agricultural research farm-a success story to emulate Lead Paper presented in National Seminar on Gender issues women in agriculture and management TNAU,Coimbatore,Tamilnadu August 20 to 22, 2001 pp 133-134 abstr.no. 237.

Kalpana Sastry, R., and Chattopadhyay, C. 2001. Fusarium wilt resistant genotype developed in safflower (Carthamus tinctorius L.) J. Mycol. Plant Pathol. 31: (in press).

Katyal, J. C., Rao, N. H., and **Reddy, M. N.** 2001. Critical aspects of organic matter management in the tropics – Example India. Journal of Nutrient Cycling in Agroecosystems, 61, 1-2, 77-88.

Kiresur, V., **Rama Rao, D.**, and **Kalpana Sastry, R.** 2001. Decision support system (DSS) in forecsting of future oilseeds scenario in India-A System Dynamic model Paper presented in First National Conference on Agro-Informatics June 3-4, 2001, UAS, Dharwad, ThemeVI: 26-27.

Mandal, U. K., Sarma, K. S. S., Victor, U. S., and **Rao, N. H.** 2001. Profile water balance under irrigated and rainfed systems, Agronomy Journal (in press).

Rama Rao, D., KalpanaSastry,R., and Soam, S. K. 2001. Training needs assessment of Indian Council of Agricultural Research Head Quarter employees. National Academy of Agricultural Research Management, Hyderabad, 25p.

Rao, K.H. and Singh, S. 2001. Lipolytic changes in cheddar cheese from buffalo milk supplemented with goat milk and ripening enzyme, Indian Journal of Dairy Science. 54 (5): 247-251.

Rao, K.H. 2001. Recent Developments in Dehydration Technology: Freeze Drying of Dairy and Food Products. In: XIII compendium on 'Advances in Preservation of Dairy and Food Products', National Dairy Research Institute, Karnal: 69-73.

Rao, N. H. 2001. Sustainable agriculture: concepts and analytical framework, Andhra Pradesh, Vision 2020: Strategy for Sustainable Agricultural Growth (Eds Raju,B.Y. and Rao,N.G.P), New Age International (P) Limited , 1-23.

Rao, N. H. 2001. Irrigation management for productive and sustainable agriculture – meeting the goals of Andhra pradesh Vision 2020: Strategy for Sustainable Agricultural Growth (Eds Raju,B.Y. and Rao,N.G.P), New Age International (P) Limited. 53-61.

Rao, V.H.P., Dharmendra, B.V., **Rao, K.H**. and Singh, R.R.B. 2001. Developments and Prospects of Osmatic Dehydration in Dairy and Food Industry, Indian Dairyman. 53 (4): 55-65.

Reddy, M. N. and **Rao, N. H**. 2002. GIS based decision support systems in agriculture, GIS India (in press).

Satyanarayana Rao, R.V. 2002. Method of screening for resistance to leafminer *Aproaerema modicella* (Deventer) under greenhouse conditions. Annals of Plant Protection Science. March 2002(10) p:38-41.

Satyanarayana Rao, R.V. 2001. Effect of biotic and abiotic factors on oviposition by the groundnut leafminer *Aproaerema modicella* (Deventer) in groundnut. Shashpa 8(2): 109-117.

Satyanarayana Rao, R.V., and Phokela, A. 2001. Joint action of cypermethrin and profenofos against Helicoverpa armigera. Presticide Research Journal 13(2): 160-164.

Soam, S. K. 2001. A case study of extension services for organic farming in UK: Options for private extension in India. In Private Extension: Indian Experiences, P. Chandra Shekara (ed.), MANAGE, Hyderabad, pp 211-221

Vidyasagar Rao, K., reviewed the project proposal on "Grape Germplasm Data base Information System" and made various suggestions for improvement as per the request made by the Director, NRC for Grapes, Pune.

Vidyasagar Rao, K., reviewed the paper "Influence of Meterogical Factors on Brown Spot Disease Development in FCV Tobacco Crop of Karnataka Light Soils" for statistical soundness and validity.

Books / Reports

- *Communication Directory of NARS*, 2nd edition, ISNAR-ICAR/NAARM project by Jagannadham Challa.
- *Focusing Agricultural Research on Poverty alleviation*, Pilot Distance Training Module, ISNAR-ICAR/ NAARM research project on Distance Training in Agricultural Research Management by Jagannadham Challa, S.K. Soam, B. S.Sontakki and V.K.J.R. Rao.

The Academy also published three bulletins viz. 'NAARM Achievements Profile', 'AHRD Project Document', and 'Rajbasha Karyanvayan Pachees Varsh Ki Pragathi Sameeksha' during the year.

Resource Papers (developed and used as input in training)

Performance appraisal and assessment of agricultural researchers (R.V. Satyanarayana Rao)
Occupational hazards in agricultural research (R.V.Satyanarayana Rao)
Recent advances in instructional technology (R.V. Satyanarayana Rao)
Motivation in work environment (R.V.Satyanarayana Rao)
Participatory learning in agricultural education- strategies and options (S.K. Soam)
Generating research questions and formulation of testable hypotheses (S.K. Soam)
Project logical framework (S.K. Soam)
Disaster management (S.K. Soam)
GIS in grazing management (S.K. Soam)
Towards sustainable livestock management : Case study for pilot distance training programme (S.K. Soam)
Guidelines for field experience training (N. Sandhya Shenoy, V.K.J. Rao, S.K. Soam, B.S. Sontakki)
Micro-level priority setting (B.S. Chandel)
Experimental designs in plant protection research (K. Vidyasagar Rao)
Notes on statistical data analysis using MSTATC and SPSS (K. Vidyasagar Rao)
Design and sampling for social research (K. Vidyasagar Rao)
Basics of statistics (K. Vidyasagar Rao)
Time management (P. Manikandan)
Goal setting (P. Manikandan)
Decision making (P. Manikandan)
Leadership (P. Manikandan)
Motivation (P. Manikandan)
Interpersonal relationship (P. Manikandan)
Biotechnological advances in animal sciences (K.H. Rao)
Environmental implications of agricultural research (K.H. Rao)
Crop-livestock interactions (K.H. Rao)
Participatory approach in agricultural research project management (B.S. Sontakki)
Instructional climate for optimizing teaching-learning effectiveness (B.S. Sontakki)
Communication strategies for personnel and organizational effectiveness (B.S. Sontakki)

Higher Degrees Awarded

Dr P. Manikandan, Principal Scientist, was awarded Master of Business Administration (MBA) degree with specialization in Human Resource Management, by the Indira Gandhi National Open University (IGNOU), New Delhi.

Dr M.M. Anwer, Principal Scientist, was awarded Postgraduate Diploma in Distance Education (PGDDE) by the Indira Gandhi National Open University (IGNOU), New Delhi.

Dr B.S. Chandel, Senior Scientist, was awarded Postgraduate Diploma in Management (PGDIM) by Indira Gandhi National Open University (IGNOU), New Delhi.

Distinguished Visitors

Dr Byron Mook, Project Manager, ISNAR, The Hague, The Netherlands, visited the Academy during May, June, and October 2001 in connection with the project on distance training for agricultural research management .

Dr Hugo Basemar of Wageningen University, The Netherlands visited the Academy during June 22-23, 2001 and interacted with the Director and faculty members on the topic of IT in scientific information.

Dr Willian Dar, Director General, ICRISAT, visited the Academy on July 10, 2001 to address the FOCARS probationers and faculty on the theme of "Science with a Human Face"

Three IAS Probationers of 2000 batch allotted to Andhra Pradesh visited the Academy on August 7, 2001.

A Study Group-I of the Parliamentary Standing Committee on Agriculture, comprising Hon'ble Members of Parliament Shri Raj Shekhar Murthy (Chairman), Shri Thawar Chand Gehlot, Shri Ramdas Rupala Gavit, Shri Y.G. Mahajan, Shri Savshibhai Makwana, Shri G. Sukender Reddy, Shri N.R.K. Reddy, Shri Chattrapal Singh, Shri A. R. Kidwai, Shri Yadlapati Venkat Rao, and Shri Dalpat Singh Paraste visited the Academy on September 21, 2001and interacted with the Director and the faculty members. The group also visited the exhibition arranged by the various ICAR institutes located in Hyderabad.

Dr S.L. Mehta, (Ex-DDG), QRT Member, and Dr H.S. Nainawatee, ADG (AHRD), and QRT Member, visited the Academy on September 22, 2001 and interacted with the Director and faculty members on the matters related to organizational structure, physical facilities, financial matters, human resources of the Academy.

Dr K.V. Raman, QRT Member, visited the Academy twice i.e.from September 26 to 29, and October 17 to 23, 2001and interacted with the Director and faculty members on ongoing and proposed further research activities of the Academy.

Dr K. Pradhan, QRT Member, visited the Academy from October 10 to 14, 2001 and interacted with the Director and faculty members on the matters related to linkages and other collaborative efforts of the Academy.

Dr Krishna Alluri and Dr Helen Lentell, Commonwealth of Learning, Canada, visited the Academy from October 22 to 26, 2001 as experts to conduct the workshop on "Modalities of implementation of pilot distance training programme under the ISNAR-ICAR/NAARM collaborative project on distance training for agricultural research management.

Dr Jocelyn Dusebio, Director, Crops Research Division, IRRI, Philippine, visited the Academy on November 29, 2001. She called on the Director, NAARM and held discussions.

A four-member delegation led by H.E. Mr Tohi Ostonsev, Director of Project Management Unit, Farm Privatization Support Project and Rural Infrastructure Rehabilitation Project, Ministry of Agriculture, Republic of Tajikistan visited the Academy on January 9, 2002.

Dr. S. B. Dandin, Direcctor, CSR&IT, Mysore, Central Silk Board visited the Academy on January 30, 2002.

The World Bank Team along with National Agricultural Technology Project (NATP) officials visited the Academy on March 9, 2002 to review the progress of two sub-projects currently in operation at the Academy and to finalize the work plan for 2002-03. The team comprising Eija Pehu, World Bank; Robert Moss, World Bank; Ashok Seth, World Bank Consultant; D.P. Singh, National Coordinator, NATP; H.C. Pathak, Director, NATP; and Rastogi, Finance Manager, NATP visited the Academy.

A three-member delegation led by Dr Asad Musayav, from Azerbaijan and Armenia visited the Academy on March 16, 2002.

Dr M. Shripathy Rao, Director, Sahakara Rural Development Academy, DCC Bank Ltd., Bidar and Shri Narayana Rao Manhalli, Vice President, DCC Bank, Ltd, Bidar visited the Academy on March 21, 2002 and held discussions with Director, NAARM.

Personnel

Director's Office

J.C. Katyal, Director (up to Jan. 7, 2002)
T. Balaguru, Acting Director (from Jan.16, 2002)
T.V. Ramdas, Personal Assistant
T. Vanisri, Personal Assistant

Joint Director's Office

S.N. Saha, Joint Director (upto Oct. 13, 2001)
Sarada Samanta, Personal Assistant

Agricultural Research Systems Management and Policies Division

T. Balaguru, Principal Scientist and Head
K.P.C. Rao, Principal Scientist (up to Feb. 15, 2002)
S. Shanmugam, Principal Scientist
K.M. Reddy, Principal Scientist
M.M. Anwar, Principal Scientist
N. Hanumantha Rao, Principal Scientist
Janardan Jee, Principal Scientist (up to Dec. 6, 2001)
R. Kalpana Sastry, Senior Scientist
B.S. Chandel, Senior Scientist
S.K. Nanda, Senior Scientist
S.K. Soam, Senior Scientist

Human Resource Development Division

P. Manikandan, Principal Scientist and Head
A. Gopalam, Principal Scientist
Jagannadham Challa, Principal Scientist
K. Hanumantha Rao, Senior Scientist
R.V.S.Rao, Senior Scientist

Information and Communication Management Division

D. Rama Rao, Principal Scientist and Head
VE. Sabarathnam, Principal Scientist and Head (up to May 31, 2001)
K. Vidyasagar Rao, Principal Scientist
M. Narayana Reddy, Principal Scientist
N. Sandhya Shenoy, Senior Scientist
B.S. Sontakki, Senior Scientist
V.K.J.R. Rao, Senior Scientist

Training Cell

S. N. Saha, In-charge
P. Vijender Reddy, Technical Officer

Administration & Finance

M. Suresh Kumar, CAO
S.K. Pathak, F&AO
K. Janakiramaiah, AAO
Y. Shankar Rao, AAO

Computer Section
M.N. Reddy, In-Charge (ARIS lab)
N.H. Rao, In-charge (Computer lab)
M.J. Pradeep Kumar, Technical Officer
K.V. Kumar, Technical Officer

Farm Services
R. Kalpana Sastry, In-Charge (up to Mar. 3, 2002)
V. Murali, In-Charge (from Mar. 4, 2002)
M.A. Basith, Technical Officer

Health Centre
A. Debnath, Medical Officer

Hostel Services
S. Shanmugam, In-Charge (up to Oct. 2, 2002)
Zameer Ahmed, Manager (Hostel Services)

Library Services
P. Manikandan, In-Charge
P.V. Nirmala, Technical Officer

Maintenance Section
V. Murali, In-Charge
Sohail Ahmed Khan, Technical Officer

Official Language Cell
A. Gopalam, In-Charge
D. Venkateswarlu, Technical Officer
J. Renuka, Asst. Director (OL)
S. Pradeep Singh, Asst. Director (OL)

Photography Section
N. Sandhya Shenoy, In-Charge
L. Venkateswarlu, Technical Officer

Press and Publication Services
D. Rama Rao, In-Charge
R.V.V.S. Prakasa Rao, Editor
P. Namdev, Technical Officer

Security Cell
B.Ch. Satyanarayana, Estate-cum-Security Officer

Transport Services
S.K. Nanda, In-Charge (up to Mar. 3, 2002)
N.R. Nageswara Rao, Technical Officer (up to Mar. 3, 2002)

Video Services
Janardan Jee, Principal Scientist (up to Dec. 6, 2001)
P. Manikandan, In-Charge (from Mar. 5,2002)
K.R. Prabhakar, Programme Officer (CCTV)
Ch. Janardhan Rao, Technical Officer

Transfers

Mr M. Suresh Kumar joined as Chief Administrative Officer on April 28, 2001 on transfer from NDRI, Karnal.

Mr S.K. Pathak, joined the Academy as FAO on July 3, 2001 on transfer from CIFE, Mumbai.

New appointments

Dr T. Balaguru, Principal Scientist, NAARM, appointed as Head, Agricultural Research Systems Management and Policies Division, on January 15, 2002.

Dr P. Manikandan, Principal Scientist, NAARM, appointed as Head, Human Resource Development Division, on January 15, 2002.

Dr D. Rama Rao, Principal Scientist, NAARM, appointed as Head, Information and Communication Management Division, on January 15, 2002.

Promotions

Name	From	To
K.R. Prabhakar	T-8	T-9
A. Debnath	T-6	T-7
Ch. Janardhan Rao	T-5	T-6
M.A. Basith	T-5	T-6
D. Venkateswarlu	T-5	T-6
Mr Bansidhar Nayak	T-4	T-5
Ahire Laxman Mahru	T-3	T-4
L. Ramesh	T-3	T-4
N, Naresh Kumar	T-3	T-4
M. Sekhar Reddy	T-3	T-4
S.K. Peera	T-3	T-4
Sham Bahadur	T-3	T-4
Savithri Murali	T-3	T-4
K. Obulpathi	T-2	T-3
M.K. Shamshuddin	T-2	T-3
M.K. Sonkusare	T-2	T-3
B.S.N. Murty	T-2	T-3
M. Mohan Rao	T-2	T-3
K. Shivaiah	T-2	T-3
Mr B. Satyanarayana	T-2	T-3
Mr N. Ashok	T-1	T-2
Mr Laxman	T-1	T-2
Mr Muthyalu	T-1	T-2
Mr U. Venkatratnam	T-1	T-2
Mr M. Padmaiah	T-1	T-2
Mr P. Eswari	T-1	T-2
Mr Prashant Gaikwad	T-1	T-2
S. Sesha Sai	Steno G.III	Personal Asst.
Ms Rajashree Bokde	LDC	UDC
Mr M. Sham Rao	SSG-I	SSG-II
Mr J. Chandraiah	SSG-I	SSG-II

Farewell

A fond farewell was given to Mr B. Pandu Reddy, Finance and Accounts Officer, of the Academy by the faculty members and staff of NAARM, on his transfer to ICAR headquarters, New Delhi. He was relieved on July 13, 2001.

Dr Janardan Jee, Principal Scientist, was relieved on December 6, 2001 from the Academy on his new assignment as Officer on Special Duty, at the newly established National Research Centre on Makhana, Darbhanga.

A fond farewell was given to Dr J.C. Katyal, Director, NAARM, by the faculty members and staff of NAARM on January 24, 2002, on his taking over as Deputy Driector General (Edn.) at ICAR, New Delhi.

Retirement

Dr VE. Sabarathnam, Principal Scientist and Head, Agricultural Research Systems Management and Policies Division retired on superannuation on May 31, 2001. NAARM family wishes a happy retired life to Dr Sabarathnam.

Dr K.P.C. Rao, Principal Scientist took voluntary retirement. He was relieved on February 15, 2002. NAARM family bids farewell to Dr Rao.

1. निष्पादन सारांश

राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी (रा.कृ.अनु.प्र.अ), जो भारतीय कृषि अनुसंधान परिषद की एक अग्रणी संस्था है। इसे सौंपे गये कृषि अनुसंधान एवं शिक्षा में उच्च स्तरीय व्यवसायिक प्रबंधन कुशलता के उत्तरदायित्व निभाने हेतु राष्ट्रीय कृषि अनुसंधान पद्धति (एन.ए.आर.एस) को ढ़ाई दशकों से अपना योगदान देती आ रही है। संरचना विकास पर बल के साथ शिक्षा, प्रशिक्षण, अनुसंधान और परामर्श के बहु–आयामी कार्यक्रम पर वर्ष 2001–2002 के दौरान अत्यधिक ध्यान दिया गया है।

इस अकादमी के वर्ष भर के रजत जयंती समारोह का आरंभ सितम्बर 01, 2001 को डॉ.एम.महादेवप्पा, अध्यक्ष, कृ.से.भ.मं, नई दिल्ली के राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी स्थापना दिवस भाषण द्वारा हुआ।

सहभागियों के पुनर्निवेशन के आधार पर बुनियादी पाठ्यक्रम में सुधार लाने के अकादमी के प्रयासों के कारण कृषि अनुसंधान सेवा बुनियादी पाठ्यक्रम का पुनर्गठन पहली बार प्रमापीय उपागम (माड्युलर अप्रोच) के रूप में किया गया है। इसका फल अधिगम परिणाम के रूप में प्राप्त हुआ। बहु–विद्याविशेष सहभागी उपागम द्वारा 'आँखों देखी पर विश्वास करना' और 'अभ्यास द्वारा सीखना' के सिद्धांतों के आधार पर क्षेत्र अनुभव प्रशिक्षण का आयोजन किया गया।

इस अकादमी के द्वारा मानव संसाधन गतिविधियों के अंतर्गत, भा.कृ.अनु.प. और राज्य कृषि विश्वविद्यालयों (रा.कृ.वि) के वैज्ञानिकों/संकाय सदस्यों के हितार्थ अनेक प्रकार के प्रशिक्षण कार्यक्रमों का आयोजन किया गया। इस रिपोर्ट की अवधि में उपर्युक्त प्रकार के 45 कार्यक्रमों का आयोजन हुआ। उनमें कुल 1028 वैज्ञानिक, संकाय, अध्यापक, प्रशासन एवं वित्त अधिकारियों ने हिस्सा लिया। इस अकादमी द्वारा दो दूरवर्ती कार्यक्रमों का भी आयोजन किया गया–आई.आई.एम. अहमदाबाद पर भा.कृ.अनु.प. के उच्चतम कार्यपालकों के लिए एक पुनश्चर्या पाठ्यक्रम (इस कार्यक्रम में अठारह वरिष्ठ स्तरीय अधिकारियों ने हिस्सा लिया) और दूसरा स्नातकोत्तर कृषि संस्थान, पेराडेनिया विश्वविद्यालय, श्रीलंका में आयोजित किया गया। (श्रीलंका के विविध अनुसंधान संगठनों के उनतालीस वैज्ञानिकों ने इस कार्यक्रम में हिस्सा लेकर लाभ उठाया)।

दो ग्रीष्मकालीन पाठशालाओं का आयोजन किया गया–एक शैक्षणिक प्रौद्योगिकी में अभिनव विकास पर था और दूसरा कृषि अनुसंधान परियोजना प्रबंध में अभिनव विकास पर। इनमें भा.कृ.अनु.प. संस्थानों और रा.कृ.वि. के कुल छियालीस वैज्ञानिकों ने हिस्सा लेकर फायदा उठाया।

कृषि मानव संसाधन विकास परियोजना के अंतर्गत, तमिलनाडु कृषि विश्वविद्यालय के आचार्य, सह–आचार्य और सहायक आचार्यों (जिनकी संख्या 162 है) को कंप्यूटर साधित शैक्षणिक प्रौद्योगिकी से सुग्राही बनाने 109 प्रशासनिक अधिकारियों को प्रशासनिक प्रबंधन और कंप्यूटर प्रयोग का प्रशिक्षण प्रदान किया गया। इस अवधि के दौरान, वरिष्ठ कार्यपालक हेतु कंप्यूटर प्रयोगशाला और कार्यपालक समिति कक्ष का विकास किया गया।

पशुचिकित्सा औषध और भेषजों के क्षेत्र में अनुसंधान एवं विकास की स्थिति के निर्धारण के लिए विज्ञान और प्रौद्योगिकी विभाग के सहयोग से 'पशुचिकित्सा औषध और भेषज' पर एक राष्ट्रीय सेमिनार का आयोजन किया गया। पद्मश्री प्रोफेसर एम.एम.शर्मा द्वारा इस सेमिनार का उद्घाटन किया गया और

इसमें भा.कृ.अनु.प. संस्थानों, रा.कृ.वि., पशुचिकित्सा औषध और भेषज उत्पादक उद्योगों और सरकारी संगठनों से 64 सहभागियों ने हिस्सा लिया।

इस वर्ष के महत्वपूर्ण उपलब्धियों में से एक है आई.एस.एन.ए.आर–आई.सी.ए.आर/एन.ए.ए.आर.एम. के सहयोग–परियोजना द्वारा 'गरीबी निर्मूलन पर कृषि अनुसंधान का केंद्रीयकरण' से संबंधित विषय पर अग्रगामी दूरस्थ प्रशिक्षण मॉड्यूल का विकास और विमोचन। इसका कार्यान्वयन पांच कृषि विश्वविद्यालयों में किया जायेगा। ये कृषि विश्वविद्यालय है–महात्मा फुले कृषि विद्यापीठ राहुरी, आचार्य एन.जी.रंगा कृषि विश्वविद्यालय, हैदराबाद, कृषि विज्ञान विश्वविद्यालय, धारवाड, कर्नाटक, तमिलनाडु कृषि विश्वविद्यालय, कोयंबतूर, टी.ए.एन.यु.वी.ए.एस, चेन्नई। पूरे देश में कार्यरत संस्थानों और वैज्ञानिकों के लाभार्थ नार्म वेब साइट (एचटीटीपी: आईसीएआर/एनएएआरएम/ईआरएनईटी.आईएन) पर राष्ट्रीय कृषि अनुसंधान पद्धति की आनलाईन डायरेक्टरी का विकास और प्रसार किया गया। यह ई–मेल, टेलीफोन, फैक्स द्वारा अनुसंधान केंद्रों से संपर्क का एक रेडीरेकनर है। भारत और विदेश में राष्ट्रीय कृषि अनुसंधान पद्धति (एन.ए.आर.एस) में कार्यरत व्यक्तियों और भारत में कृषि विकास से संबद्ध अन्य अभिकरणों के लिए यह बड़ा उपयोगी है।

एन.ए.टी.पी. उप परियोजना के अंतर्गत, 'संस्थानीय प्राथमिकता सेटिंग, मानिटरिंग और मूल्यांकन तथा समाज वैज्ञानिकों को नेटवर्किंग', नीतिगत अनुसंधान और विस्तार योजना (एस.आर.ई.पी) मार्गदर्शक में संशोधन अनुसंधान तत्व को सुदृढ बनाकर तथा संशोधित मार्गदर्शक का उपयोग प्रभावी एस.आर.ई.पी. विकसित करने में किया जा रहा है। मानव संसाधन विकास उप–परियोजना के अंतर्गत लिंग (जेंडर) पर कई प्रकार की कंप्यूटर आधारित शैक्षणिक सामग्री विकसित की गई। वर्षा आधारित धान उत्पादन में अनुसंधान की प्राथमिकता और लक्षणों के विषय अध्ययन का अभ्यास जिला स्तर के ए.आर.सी.वी आई.ई.डब्ल्यू.जी.आई.एस. डाटाबेस उपयोग कर विकसित किया गया।

आंध्रप्रदेश सेस फण्ड अनुसंधान योजना के तहत, 'कृषि पाठ्यक्रम में लिंग (जेंडर) हेतु शिक्षण मॉड्युल विकसित करना'। इसमें लिंग–भूमिका, लिंग अनुसंधान प्रबंध और लिंग समय प्रबंध पर तीन शिक्षण माड्यूल के मुद्रण के साथ सी.डी. भी तैयार की गई। विकसित मॉड्यूल टी.एन.ए.यु. कोयंबतूर और ए.एन.जी.आर.ए.यु. हैदराबाद के शिक्षकों के समक्ष प्रस्तुत किए गए।

भारत में फार्मिंग सिस्टम रिसर्च (एफ.एस.आर) कार्यान्वयन उपागम को जानने के लिए कृषि अनुसंधान स्टेशन, अरूपकोट्टई टी.एन.ए.यू. कोयंबतूर द्वारा इंटिग्रेटेड फार्मिंग सिस्टम डेवलपमेंट प्रोग्राम के प्राथमिक और द्वितीय स्तर पर संचयित सूचना के आधार पर विषय अध्ययन विकसित किया गया। वैज्ञानिकों द्वारा विकसित संघटित फार्मिंग पद्धति की उत्पादकता, लाभ और स्थायित्व का अनुसंधान केंद्र के साथ–साथ कृषकों के खेत में परीक्षित पद्धति का विषय अध्ययन में विश्लेषण कर के व्याख्या की गई।

कृषि अनुसंधान और शिक्षण पद्धति में प्रबंध से संबद्ध अकादमी द्वारा भा.कृ.अनु.प. को दिए नीति समर्थन के तहत–एक दिवसीय बैठक का आयोजन भा.कृ.अनु.प. के वैज्ञानिकों के कार्य निष्पादन की समीक्षा और परियोजना आधारित बजटिंग के पुष्टीकरण हेतु नई प्रणाली तैयार करने के लिए किया गया। अकादमी द्वारा भा.कृ.अनु.प. कर्मचारियों की प्रशिक्षण आवश्यकताओं का निर्धारण विकसित किया गया है जिसके अनुमोदन के पश्चात इसे कार्यान्वित भी किया गया है।

अकादमी द्वारा चावल अनुसंधान निदेशालय, हैदराबाद के संकर चावल फसल उत्पादन पर वीडियो कार्यक्रम तथा राष्ट्रीय कृषि वानिकी अनुसंधान केंद्र, झाँसी पर एक वीडियो फिल्म के निर्माण का अत्यधिक काम पूरा किया गया है।

राष्ट्रीय कृषि प्रौद्योगिकी परियोजना के तहत ओहियो राज्य विश्वविद्यालय, संयुक्त राज्य अमेरिका में मानव संसाधन विकास में एक संकाय सदस्य ने तीन महिनों का उन्नत प्रशिक्षण तथा एक अन्य संकाय सदस्य ने इस्नार, नैदरलैण्ड में कृषि अनुसंधान संगठनों में निष्पादन मूल्यांकन पर तीन सप्ताह का अध्ययन दौरा किया ।

अकादमी के पुस्तकालय में पहली ही बार डिजिटैजेशन किया जा रहा है। पुस्तकालय की गतिविधियों की कंप्यूटरीकरण करने हेतु तथा पुस्तकालय संसाधनों के ओ.पी.ए.सी. वेब के सृजन के लिए लिब एस.वाई.एस. पैकेज खरीदा गया। एक छोटा सर्वर भी खरीदा गया जो उपलब्ध लिब एस.वाई.एस. पैकेज के प्रभावी प्रयोग हेतु उपयोगी सिद्ध होगा। संकाय सदस्यों ने विभिन्न कार्यशाला तथा संगोष्ठियों में 28 वैज्ञानिक लेख तथा 23 अन्य लेख प्रस्तुत किए। इस वर्ष के अन्तर्गत, तीन बुलेटिन जैसे 'नार्म की उपलब्धियों की रूपरेखा' 'कृषि मानव संसाधन परियोजना विकास प्रलेख' तथा 'राजभाषा कार्यान्वयन : पच्चीस वर्ष की प्रगति समीक्षा' का विमोचन किया गया।

अकादमी को नगर राजभाषा कार्यान्वयन समिति, दक्षिण मध्य रेलवे द्वारा 1999–2000 वर्ष के लिए नगरद्वय राजभाषा कार्यान्वयन में उत्तम कार्य निष्पादन हेतु 'राजभाषा पुरस्कार' से सम्मानित किया गया।

डॉ.जे.सी.कत्याल, निदेशक को कृषि में सूक्ष्म पोषक तत्व पर शोध कार्य के लिए एस.एस.रानडे स्मारक ट्रस्ट का 'लाईफ टाईम अचीवमेंट अवार्ड' से सम्मानित किया गया। नार्म खेल दल को भारतीय कृषि अनुसंधान संस्थान, नई दिल्ली में आयोजित फाइनल अन्तर क्षेत्रीय प्रतियोगिता तथा केंद्रीय मत्स्य प्रौद्योगिकी संस्थान में आयोजित भा.कृ.अनु.प. अन्तर संस्थानीय प्रतियोगिता में एक दल ईवेंट के लिए तथा पाँच व्यक्तिगत पुरस्कार प्राप्त हुए। अकादमी को पाँच व्यक्तिगत पुरस्कारों के साथ–साथ महिलाओं के लिए 'ओवराल चैंपियनशीप' से भी सुशोभित किया गया।

अकादमी को बागबानी केंद्र हैदराबाद में आयोजित 26 वें वार्षिक गुलाब प्रदर्शन में हैदराबाद रोज सोसायटी चैलेन्ज कप, दक्कन क्रानिकल ट्राफी एवं टी.चंद्रशेखर रेड्डी मेमोरियल ट्राफी जैसे शोभायमान पुरस्कार प्राप्त हुए।

नार्म की गतिविधियों की एक झलक

अप्रैल 2001

* बीज विज्ञान एवं प्रौद्योगिकी के विद्या–विषय में स्नातकोत्तर स्तर पर पाठ्यक्रम में एकरूपता लाने हेतु 6–7 अप्रैल, 2001 के दौरान एक बैठक व कार्यशाला आयोजित की गई। इस कार्यक्रम में 34 सदस्यों ने भाग लिया। डॉ.एस.एन.साहा, डॉ.ए.गोपालम एवं डॉ.हनुमंत राव ने इस कार्यशाला का समन्वयन किया।

* 10 से 12 अप्रैल, 2001 के दौरान अभिप्रेरणा तकनीक पर तीन दिवसीय कार्यशाला आयोजित की गई है। इस कार्यशाला में चार सदस्यों ने भाग लिया। डॉ.एम.एम.अनवर एवं डॉ.जनार्दन जी ने इस कार्यशाला का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय के शिक्षकों हेतु कंप्यूटर युक्त शैक्षणिक प्रौद्योगिकी पर जागृतिक प्रशिक्षण कार्यक्रम 10 से 12 अप्रैल, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में तमिलनाडु कृषि विश्वविद्यालय के 18 शिक्षकों ने भाग लिया। डॉ.ए.गोपालम, डॉ.एस.के.सोम एवं डॉ.आर.वी.एस.राव ने कार्यक्रम का समन्वयन किया।

* अप्रैल 16 से 21, 2001 के दौरान वेबपेजस की अभिकल्पना एवं विकास पर एक सप्ताह का प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में कृषि प्रौद्योगिकी सूचना केंद्रों के 12 वैज्ञानिकों ने भाग लिया। डॉ.एन.संध्या शेणायें, डॉ.एम.एन.रेड्डी ने इस कार्यक्रम का समन्वयन किया।

* प्रशिक्षकों के प्रशिक्षण पर 17 से 28 अप्रैल, 2001 के दौरान दो सप्ताह का कार्यक्रम आयोजित किया गया। डॉ.टी.बालगुरू एवं डॉ.पी.मणिकण्ठन ने इस कार्यक्रम का समन्वयन किया।

* कृषि अनुसंधान परियोजना प्रबंध में अद्यतन उन्नतियों पर 18 अप्रैल से 8 मई, 2001 के दौरान एक ग्रीष्मकालीन विद्यालय चलाया गया। इस कार्यक्रम में 21 सहभागियों ने भाग लिया। डॉ.जे.सी.कत्याल इस ग्रीष्मकालीन विद्यालय के निदेशक रहे। डॉ.टी.बालगुरू , डॉ.आर.कल्पना शास्त्री, डॉ.वी.के.जयराघवेंद्रराव एवं भरत एस. सोनटकी ने इस कार्यक्रम का समन्वयन किया।

* कृषि अनुसंधान में पाँचवा प्रबंध विकास कार्यक्रम 19–25 अप्रैल, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के 6 विभागों के प्रमुखों ने भाग लिया। डॉ.जगन्नाथम चल्ला एवं डॉ.एस.एन.साहा ने इस कार्यक्रम का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय संकाय के लिए कंप्यूटर आधारित शैक्षणिक प्रौद्योगिकी पर 26 अप्रैल से 02 मई, 2001 के दौरान कार्यक्रम आयोजित किया गया। इस कार्यक्रम में तमिलनाडु कृषि विश्वविद्यालय के 21 संकाय सदस्यों ने भाग लिया। डॉ.के.विद्यासागर राव, डॉ.एन.एन.अनवर एवं डॉ.आर.कल्पना शास्त्री ने इस कार्यक्रम का समन्वयन किया।

मई 2001

* 10–30 मई, 2001 के दौरान शैक्षणिक प्रौद्योगिकी में अद्यतन उन्नतियों पर तीन सप्ताह का ग्रीष्मकालीन विद्यालय आयोजित किया गया। इस कार्यक्रम में राज्य कृषि विद्यालयों के 25 सहायक प्राध्यापकों एवं सह–प्राध्यापकों ने भाग लिया। डॉ.ए.गोपालम ने इस ग्रीष्मकालीन विद्यालय का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय, कोयंबत्तूर के प्रशासनिक एवं लेखा अधिकारियों के लिए 17–19 मई, 2001 के दौरान प्रशासन के आधुनिक प्रबंध पर तीन दिवसीय प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में 26 सदस्यों ने भाग लिया। श्री एम.सुरेश कुमार, श्री बी.पाण्डु रेड्डी एवं डॉ.एन.संध्या शेणायॅ ने इस कार्यक्रम का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय के प्राध्यापकों के लिए 28–30 मई, 2001 के दौरान कंप्यूटर युक्त शैक्षणिक प्रौद्योगिकी पर तीन दिवसीय सुग्राह्य प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में 28 सदस्यों ने भाग लिया। डॉ.टी.बालगुरू, डॉ.एस.के.नन्दा एवं डॉ.बी.एस.सोनटक्की ने इस कार्यक्रम का समन्वयन किया।

* प्रौद्योगिकी पूर्वानुमान एवं निर्धारण परिषद के पेटेंट समन्वयन केंद्र (टी.आई.एम.ए.सी.), विज्ञान एवं प्रौद्योगिकी विभाग, नई दिल्ली के सौजन्य से 31 मई, 2001 के दौरान भारतीय कृषि में बौद्धिक संपदा अधिकार पर एक दिवसीय जागृतिका कार्यशाला आयोजित की गई। इस कार्यशाला में 63 वैज्ञानिकों ने भाग लिया। डॉ.आर.कल्पना शास्त्री ने इस कार्यशाला का समन्वयन किया।

जून 2001

* वैयक्तिक एवं संगठनात्मक प्रभावात्मकता हेतु कृषि वैज्ञानिक विकास पर दस दिवसीय कार्यक्रम 12–23 जून, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के आठ वरिष्ठ वैज्ञानिकों ने भाग लिया। डॉ.एस.शणमुगम एवं डॉ.जगन्नाथम चल्ला ने इस कार्यक्रम का समन्वयन किया।

* 18–23 जून, 2001 के दौरान प्रभावी प्रस्तुतीकरण के लिए शैक्षणिक उपकरण पर एक सप्ताह का प्रशिक्षण कार्यक्रम आयोजित किया गया। डॉ.एन.संध्या शेणायॅ एवं डॉ.आर.कल्पना शास्त्री ने इस कार्यक्रम का समन्वयन किया।

* कृषि अनुसंधान प्राथमिकता तकनीक पर 25–30 जून, 2001 के दौरान एक सप्ताह का प्रशिक्षण कार्यक्रम आयोजित किया गया। इस कार्यक्रम में 27 वैज्ञानिकों ने भाग लिया। डॉ.के.पी.सी.राव एवं डॉ.बी.एस.चन्देल ने इस कार्यक्रम का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय के संकाय सदस्यों के लिए कंप्यूटर आधारित शैक्षणिक प्रौद्योगिकी में एक सप्ताह का संकाय विकास कार्यक्रम 26 जून से 02 जुलाई, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में 32 संकाय सदस्यों ने भाग लिया। डॉ.एन.हनुमंत राव एवं डॉ.के.एम.रेड्डी ने इस कार्यक्रम का समन्वयन किया।

जुलाई 2001

* गहन हिन्दी प्रशिक्षण 2–6 जुलाई, 2001 के दौरान पाँच दिवसीय कार्यशाला आयोजित किया गया। इस कार्यशाला में 41 सदस्यों ने भाग लिया। डॉ.ए.गोपालम एवं श्री प्रदीप सिंह ने इस कार्यशाला का समन्वयन किया।

* कृषि में सूचना प्रौद्योगिकी पर तीन सप्ताह की कार्यशाला 10–30 जुलाई, 2001 के दौरान आयोजिक की गई। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों तथा राज्य कृषि विश्वविद्यालय के 28 वैज्ञानिकों ने भाग लिया। डॉ.एन.हनुमंत राव एवं डॉ.एम.नारायण रेड्डी ने इस कार्यक्रम का समन्वयन किया।

* 23–31 जुलाई, 2001 के दौरान एन.ए.टी.पी. के तहत प्रशासनिक एवं वित्तीय प्रबंध में संगठनात्मक एवं प्रबंध सुधार पर एक सप्ताह का कार्यक्रम आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के 34 प्रशासनिक एवं वित्तीय कार्मिकों ने भाग लिया। श्री एम. सुरेश कुमार, श्री एस.के.पाठक एवं डॉ.बी.एस.सोनटक्की ने इस कार्यक्रम का समन्वयन किया।

अगस्त 2001

* कृषि संपोषणीयता के मूल्यांकन के लिए विश्लेषणात्मक औजार एवं तकनीक पर 1–10 अगस्त, 2001 के दौरान 10 दिवसीय कार्यशाला आयोजित की गई। इस कार्यक्रम में भा.कृ.अनु.प. संस्थान तथा राज्य कृषि विश्वविद्यालय के 12 वैज्ञानिकों ने भाग लिया। डॉ.एम.नारायण रेड्डी एवं डॉ.एन.हनुमंत राव ने इस कार्यक्रम का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय के प्रशासनिक स्टाफ के लिए कृषि अनुसंधान एवं शिक्षा के लिए कंप्यूटर प्रयोग पर एक सप्ताह का कार्यक्रम 2–8 अगस्त, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में 21 सदस्यों ने भाग लिया। डॉ.एम.नारायण रेड्डी, श्री एम.सुरेश कुमार एवं श्री एस.के.पाठक ने इस कार्यक्रम क समन्वयन किया।

* 20–31 अगस्त, 2001 के दौरान दो सप्ताह का कार्यक्रम आयोजित किया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के 18 वैज्ञानिकों ने भाग लिया। डॉ.के.एम.रेड्डी एवं डॉ.के.विद्यासागर राव ने इस कार्यक्रम का समन्वयन किया।

सितम्बर 2001

* कृषि अनुसंधान परियोजना प्रबंध में अद्यतन उन्नतियों पर 10–30 सितम्बर, 2001 के दौरान तीन सप्ताह का एक पुनश्चर्या पाठ्यक्रम आयोजित किया गया। इस कार्यक्रम में 19 सदस्यों ने भाग लिया। डॉ.डी.रामाराव एवं डॉ.एम.एम.अनवर ने पाठ्यक्रम का समन्वयन किया।

* कृषि अनुसंधान एवं विकास के प्रभावात्मक मूल्यांकन पर एक सप्ताह का प्रशिक्षण कार्यक्रम 17–22 सितम्बर, 2001 के दौरान आयोजित किया गया। डॉ.बी.एस.चन्देल एवं डॉ.के.पी.सी.राव इस कार्यक्रम का समन्वयन किया।

अक्तूबर 2001

* तनाव प्रबंध पर एक सप्ताह का वरिष्ठ स्तर का कार्यक्रम 17–24 अक्तूबर, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के चार वैज्ञानिकों ने भाग लिया। डॉ.एस.शणमुगम एवं डॉ.पी.मणिकण्डन इस कार्यक्रम के समन्वयक रहे।

* प्रायोगिक सुदुर पशिक्षण कार्यक्रम के कार्यान्वयन के पद्धतियों पर एक प्रशिक्षण कार्यक्रम कॉमनवेल्थ ऑफ लर्निंग, कैनडा तथा इग्नोर के मार्गदर्शन के अन्तर्गत 21–26 अक्तूबर, 2001 के दौरान आयोजित किया गया। कुल 14 अध्ययन केंद्रों के पदाधिकारी समन्वयक एवं राज्य कृषि विश्वविद्यालय के पाँच नोडल अधिकारी तथा नार्म परियोजना टीम के सहित सभी ने इसमें भाग लिया।

* कृषि अनुसंधान में जी.आई.एस. प्रयोग पर दस दिवसीय प्रशिक्षण कार्यक्रम 30 अक्तूबर से 9 नवम्बर, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में 8 सदस्यों ने भाग लिया। डॉ.एन.हनुमंत राव एवं डॉ.एम.नारायण रेड्डी में इस कार्यक्रम का समन्वयन किया।

नवम्बर, 2001

* भा.कृ.अनु.प. में राजभाषा नीति की प्राथमिकताएँ एवं आवश्यकताएँ पर चार दिवसीय कार्यशाला 6–9 नवम्बर, 2001 के दौरान आयोजित की गई। इस कार्यशाला में 47 सदस्यों ने भाग लिया। डॉ.ए.गोपालम एवं श्री डी.वेंकटेश्वर्लु ने इस कार्यशाला का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय के प्राध्यापकों के लिए कंप्यूटर आधारित शैक्षणिक, प्रौद्योगिकी पर जागरूकता प्रशिक्षण कार्यक्रम 8–10 नवम्बर, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में तमिलनाडु कृषि विश्वविद्यालय के 35 प्राध्यापकों ने भाग लिया। डॉ.पी.मणिकण्डन, डॉ.एम.नारायण रेड्डी एवं डॉ.एस.के.सोम ने इस कार्यक्रम का समन्वयन किया।

* 19–24 नवम्बर, 2001 के दौरान कृषि अनुसंधान एवं प्रबंध के लिए उन्नत डाटा विश्लेषणात्मक तकनीक पर दस दिवसीय कार्यक्रम किया गया। इस कार्यक्रम में दो सदस्यों ने भाग लिया। डॉ.एम.नारायण रेड्डी एवं डॉ.के.विद्यासागर राव ने इस कार्यक्रम का समन्वयन किया।

* तमिलनाडु कृषि विश्वविद्यालय के सहायक प्राध्यापकों एवं सह प्राध्यापकों के लिए 19–25 नवम्बर, 2001 के दौरान कंप्यूटर आधारित शैक्षणिक, प्रौद्योगिकी पर एक सप्ताह का कार्यक्रम आयोजित किया गया। इस कार्यक्रम में तमिलनाडु कृषि विश्वविद्यालय के 28 सदस्यों ने भाग लिया। डॉ.के.पी.सी.राव, डॉ.एन.हनुमंत राव एवं डॉ.ए.गोपालम ने इस कार्यक्रम का समन्वयन किया।

* नवम्बर 21–27, 2001 के दौरान कृषि अनुसंधान में प्रबंध विकास कार्यक्रम पर एक सप्ताह का कार्यक्रम आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के 14 विभागाध्यक्षों ने इसमें भाग लिया। डॉ.जगन्नाथम चल्ला एवं डॉ.आर.वी.एस.राव ने इस कार्यक्रम का समन्वयन किया।

दिसम्बर 2001

* कृषि में सूचना प्रौद्योगिकी पर एक पुनश्चर्या पाठ्यक्रम 3–23 दिसम्बर, 2001 के दौरान आयोजित किया गया। भा.कृ.अनु.प. संस्थानों तथा राज्य कृषि विश्वविद्यालयों के बीस वरिष्ठ वैज्ञानिकों ने इस कार्यक्रम में भाग लिया। डॉ.बी.एस.सोनटक्की एवं डॉ.एस.के.सोम ने कार्यक्रम का समन्वयन किया।

* कृषि में विकासोन्मुख अनुसंधान पर एक सप्ताह का कार्यक्रम 10–15 दिसम्बर, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में दस सदस्यों ने भाग लिया। डॉ.एस.के.सोम एवं डॉ.बी.एस.सोनटक्की ने इस कार्यक्रम का समन्वयन किया।

* कृषि अनुसंधान एवं शिक्षा में कंप्यूटर प्रयोग पर एक सप्ताह का प्रायोजित कार्यक्रम 10–16 दिसम्बर, 2001 के दौरान आयोजित किया गया। तमिलनाडु कृषि विश्वविद्यालय के 28 प्रशासनिक एवं वित्तीय अधिकारियों ने इस कार्यक्रम में भाग लिया। डॉ.आर.कल्पना शास्त्री, डॉ.एन.हनुमंत राव एवं डॉ.बी.एस.चन्देल ने इस कार्यक्रम का समन्वयन किया

* कृषि अनुसंधान एवं प्रबंध पर चार दिवसीय कार्यकारी विकास कार्यक्रम 21–24 दिसम्बर, 2001 के दौरान आयोजित किया गया। इस कार्यक्रम में 14 सदस्यों ने भाग लिया। डॉ.जगन्नाथम चल्ला एवं डॉ.एस.एन.साहा ने कार्यक्रम का समन्वयन किया।

* शैक्षणिक मनोविज्ञान एवं नेतृत्व विकास पर तीन दिवसीय प्रायोजित कार्यक्रम 27–29 दिसम्बर, 2001 के दौरान आयोजित किया गया। तमिलनाडु कृषि विश्वविद्यालय के कुल 54 सहायक एवं सह प्राध्यापकों ने इस कार्यक्रम में भाग लिया। डॉ.ए.गोपालम एवं डॉ.एस.एन.साहा ने इस कार्यक्रम का समन्वयन किया।

जनवरी 2002

* 16–24 जनवरी, 2002 के दौरान प्रशासनिक एवं वित्तीय प्रबंध पर एक सप्ताह का पुनश्चर्या पाठ्यक्रम आयोजित किया गया। भा.कृ.अनु.प. संस्थानों के 25 प्रशासनिक अधिकारी एवं वित्त एवं लेखा अधिकारियों ने इस कार्यक्रम में भाग लिया। श्री एम.सुरेश कुमार एवं श्री एस.के.पाठक ने इस कार्यक्रम का समन्वयन किया।

* कृषि में मानव शक्ति योजना के लिए पद्धतियाँ विषय पर तीन दिवसीय कार्यक्रम 17–19 जनवरी, 2002 के दौरान आयोजित किया गया। भा.कृ.अनु.प. संस्थान तथा राज्य कृषि विश्वविद्यालय के 14 प्राध्यापक एवं सह प्राध्यापकों ने इस कार्यक्रम में भाग लिया। डॉ.डी.रामाराव एवं डॉ.एस.के.नन्दा ने इस कार्यक्रम का समन्वयन किया।

* भा.कृ.अनु.प. के लिए मार्गदर्शी सिद्धान्त तथा नये नियमों की संरचना तथा मूल्यांकन पर 25 जनवरी, 2002 को एक दिवसीय बैठक आयोजित की गई। इस कार्यक्रम में भा.कृ.अनु.प. संस्थानों के 30 निदेशक एवं विभाग प्रमुखों ने भाग लिया। डॉ.डी.रामाराव ने इस कार्यक्रम का समन्वयन किया।

फरवरी 2002

* 4–8 फरवरी, 2002 के दौरान 'गहन हिन्दी प्रशिक्षण व कार्यशाला पाँच दिवसीय कार्यक्रम आयोजित किया गया। डॉ.ए.गोपालम एवं श्रीमती जे.रेणुका ने कार्यशाला का समन्वयन किया।

* कृषि में कंप्यूटर प्रयोग पर दस दिवसीय कार्यक्रम 5–15 फरवरी, 2002 के दौरान आयोजित किया गया। इस कार्यक्रम में 7 सदस्यों ने भाग लिया। डॉ.के.विद्यासागर राव एवं डॉ.के.एम.रेड्डी ने कार्यक्रम का समन्वयन किया।

मार्च 2002

* माइक्रोसाफ्ट परियोजना के प्रयोग से परियोजन प्रबंध पर पाँच दिवसीय कार्यक्रम 4–8 मार्च, 2002 के दौरान आयोजित किया गया। इस कार्यक्रम में 6 सदस्यों ने भाग लिया। डॉ.एस.के.नन्दा, डॉ.एन.एच.राव एवं डॉ.डी.रामाराव ने इस कार्यक्रम में समन्वयन किया।

* परिवर्तन प्रबंध पर तीन दिवसीय कार्यशाला 14–16 मार्च, 2002 के दौरान आयोजित की गई। 8 सदस्यों ने इस कार्यक्रम में भाग लिया। डॉ.एस.एन.साहा, डॉ.एम.एस.अनवर एवं डॉ.बी.एस.सोनटक्की ने कार्यक्रम का समन्वयन किया।

* विज्ञान एवं प्रौद्योगिकी विभाग, नई दिल्ली द्वारा प्रायोजित पशुचिकित्सा औषधी पर दो दिवसीय राष्ट्रीय संगोष्ठी 23–34 मार्च, 2002 के दौरान आचार्य एन.जी.रंगा कृषि विश्वविद्यालय, हैदराबाद के सहयोग से आयोजित किया गया। चिकित्सा, पशुचिकित्सा, विषविज्ञान, औषधविज्ञान के क्षेत्र के 75 वैज्ञानिक, शिक्षाविद् निजी क्षेत्र के कार्यपालकों ने इस संगोष्ठी में भाग लिया। डॉ.डी.रामाराव, नार्म एवं डॉ.के.एस.रेड्डी, आचार्य एन.जी.रंगा. कृषि विश्वविद्यालय ने संगोष्ठी का समन्वयन किया।